General Biology I

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Course: **General Biology I**
Sec.: 1
Class:11:15 - 12:15
Place: M2 203

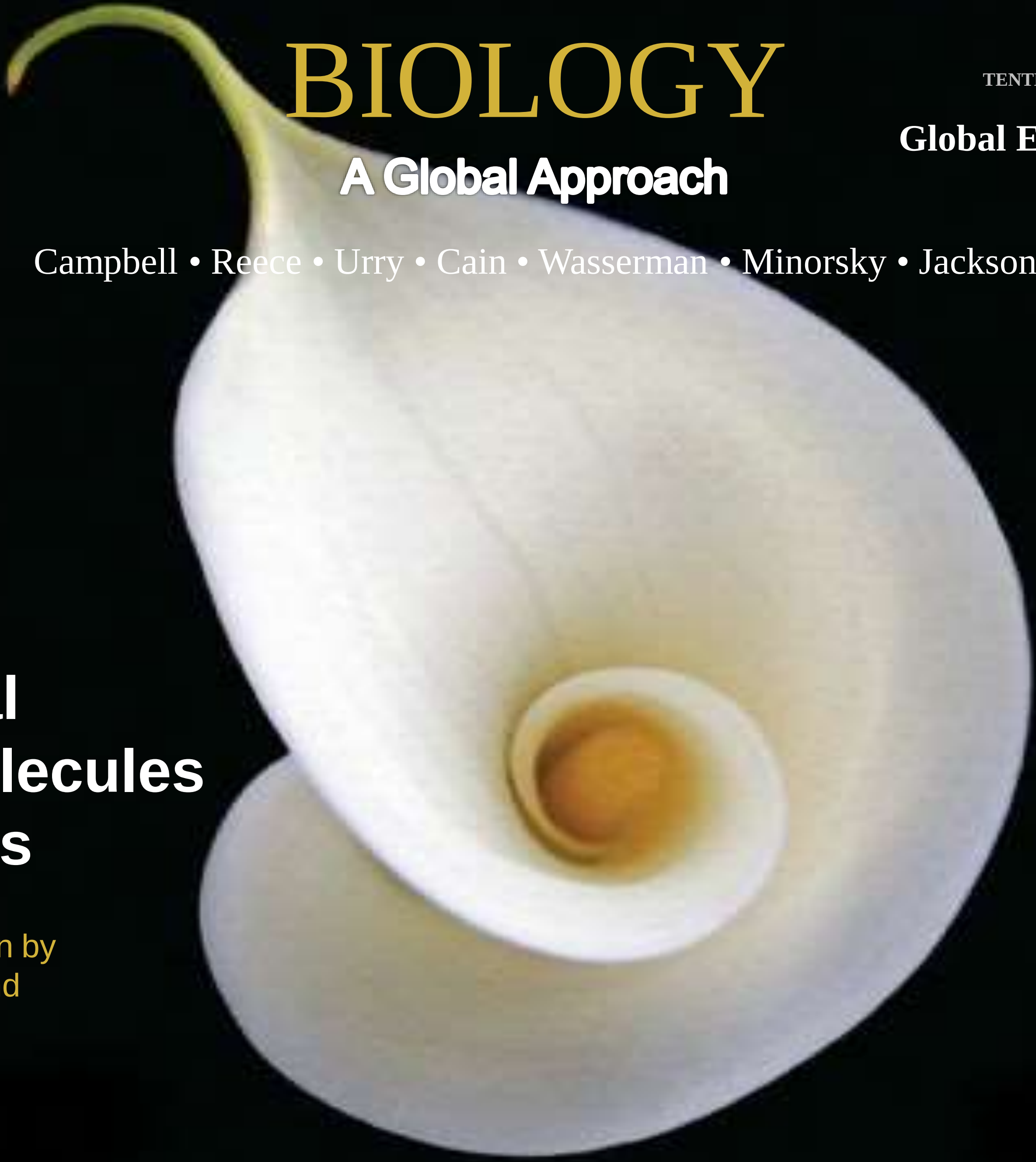
BIOLOGY

TENTH EDITION

Global Edition

A Global Approach

Campbell • Reece • Urry • Cain • Wasserman • Minorsky • Jackson



5

Biological Macromolecules and Lipids

Lecture Presentation by
Nicole Tunbridge and
Kathleen Fitzpatrick

The Molecules of Life

a) All living things are made up of four classes of large biological molecules: carbohydrates, lipids, proteins, and nucleic acids

b) Macromolecules are large molecules and are complex

c) Large biological molecules have unique properties that arise from the orderly arrangement of their atoms

Figure 5.1

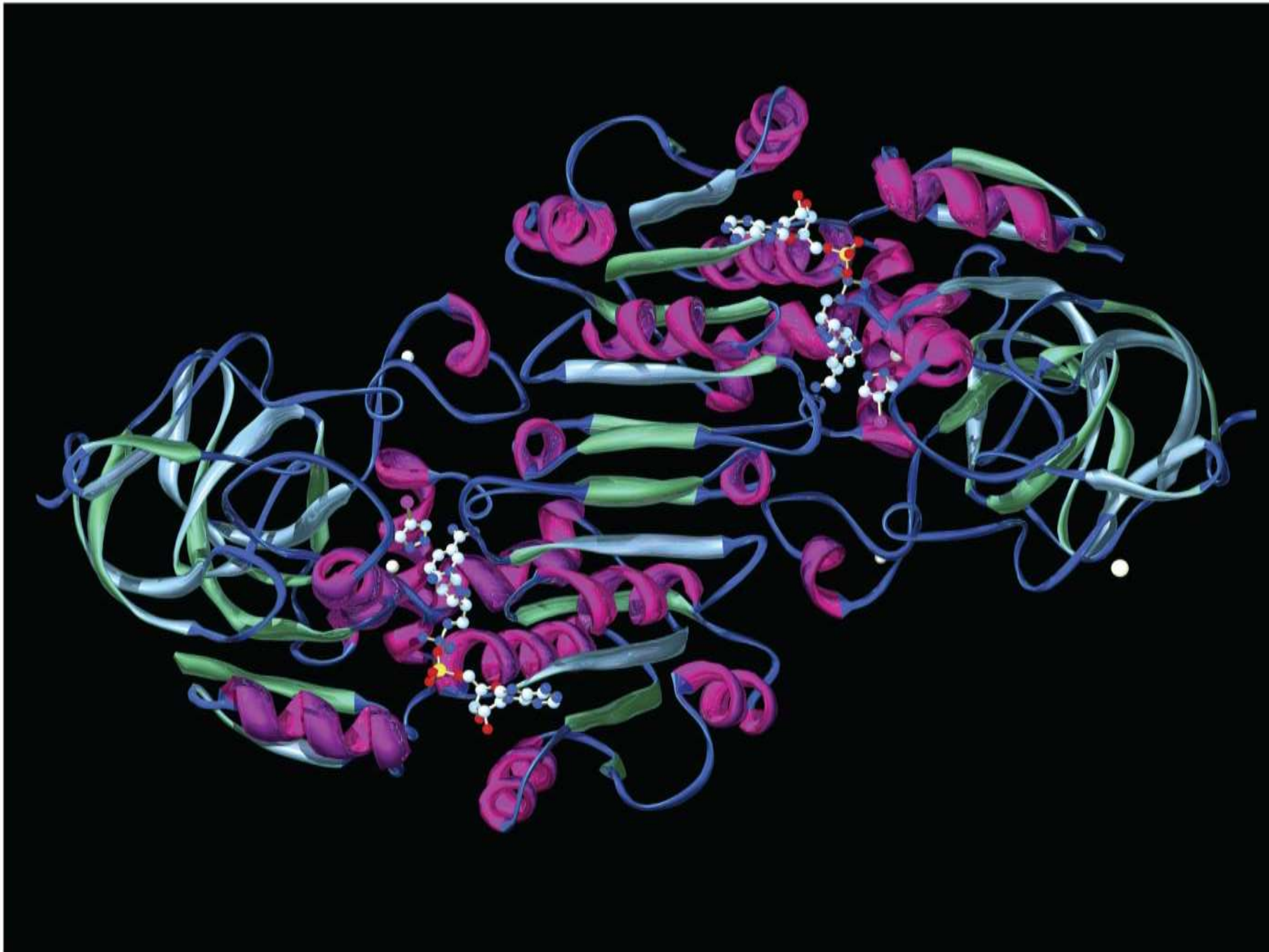
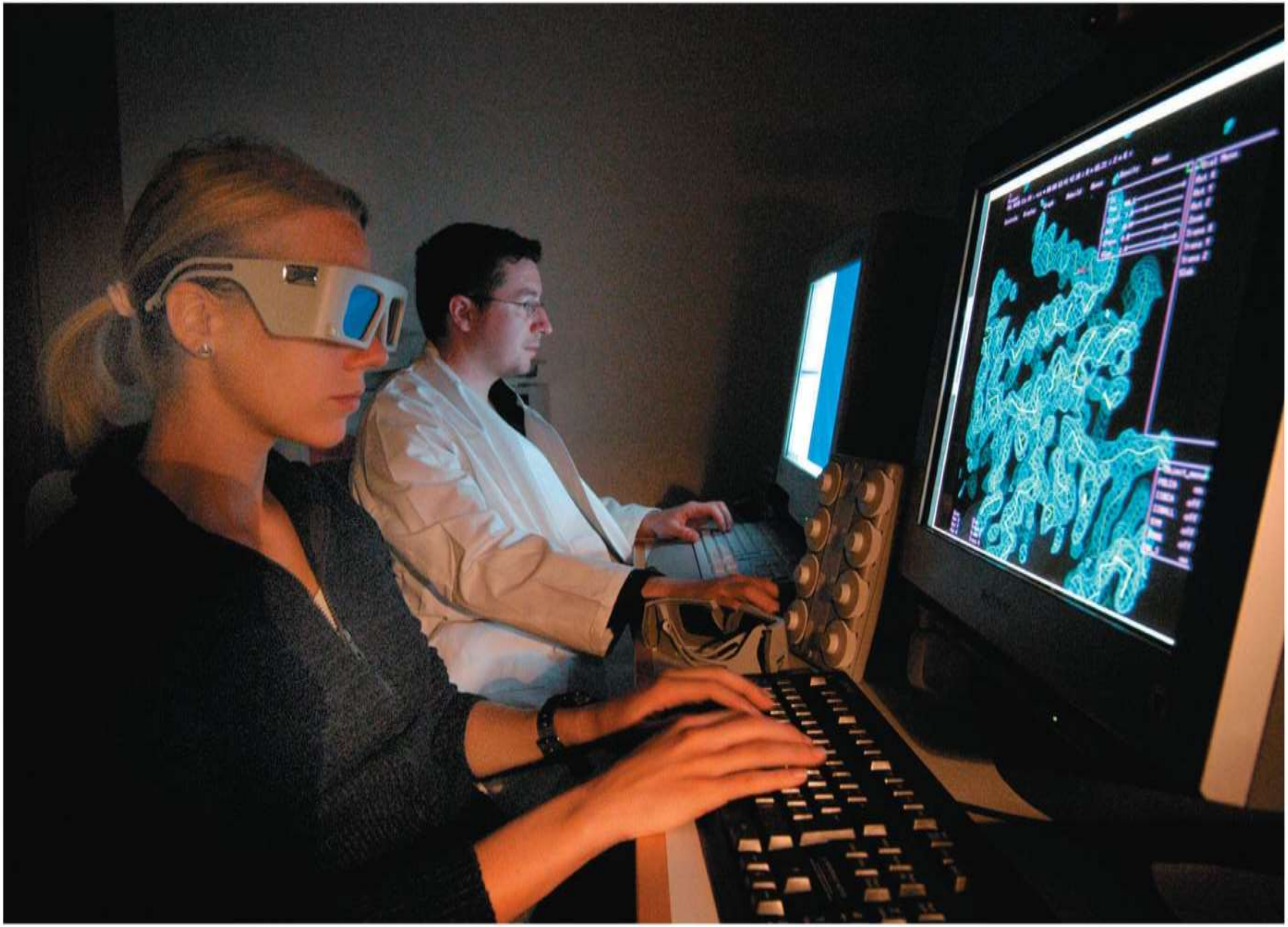


Figure 5.1a



Concept 5.1: Macromolecules are polymers, built from monomers

a) A **polymer** is a long molecule consisting of many similar building blocks

b) The repeating units that serve as building blocks are called **monomers**

c) Three of the four classes of life’s organic molecules are polymers

  a) Carbohydrates

  b) Proteins

  c) Nucleic acids

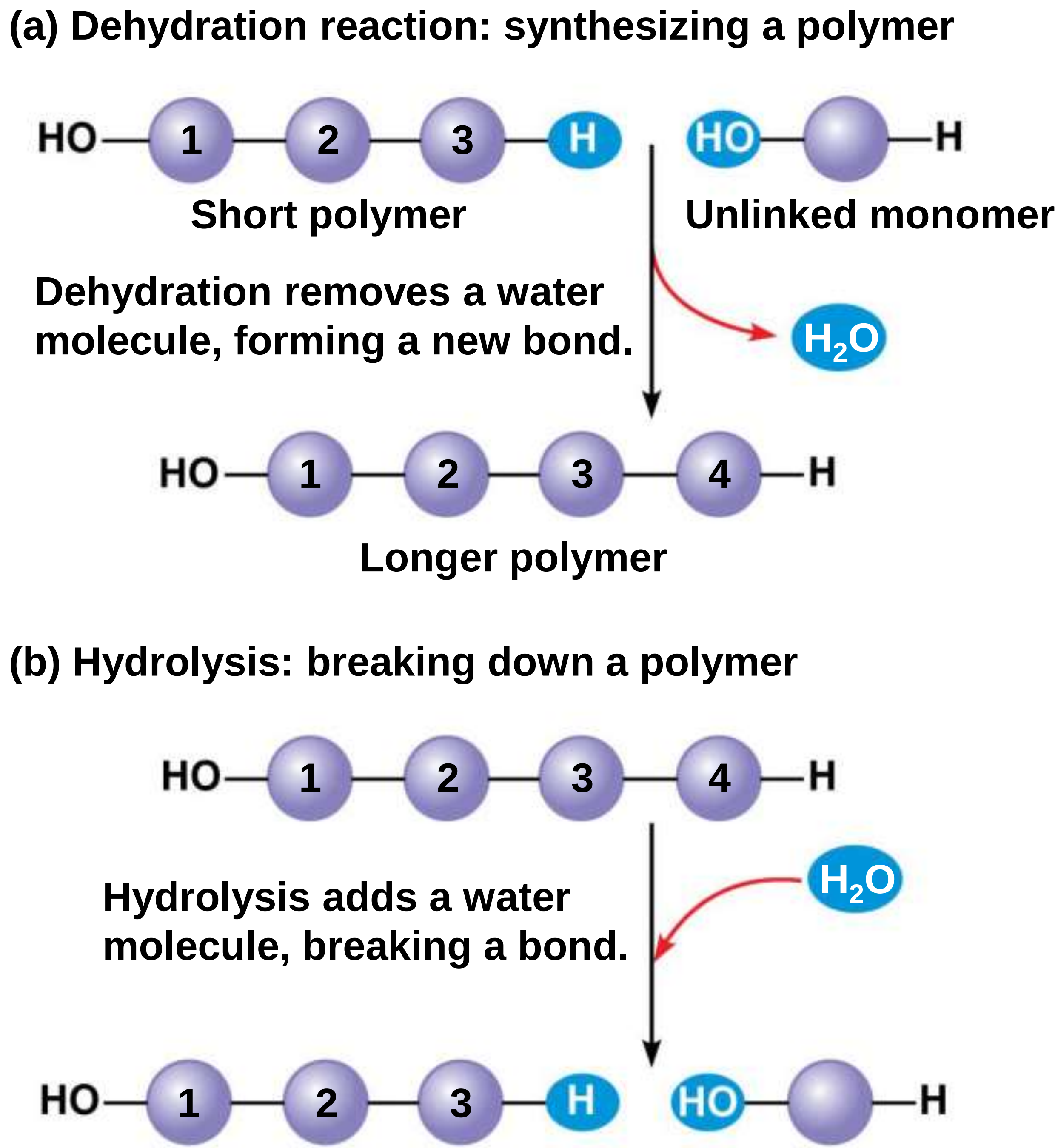
The Synthesis and Breakdown of Polymers

a) **Enzymes** are specialized macromolecules that speed up chemical reactions such as those that make or break down polymers

b) A **dehydration reaction** occurs when two monomers bond together through the loss of a water molecule

c) Polymers are disassembled to monomers by **hydrolysis**, a reaction that is essentially the reverse of the dehydration reaction

Figure 5.2


(a) Dehydration reaction: synthesizing a polymer
HO 1 2 3 H
Short polymer
HO H
Unlinked monomer
Dehydration removes a water molecule, forming a new bond.
H_2O
HO 1 2 3 4 H
Longer polymer
(b) Hydrolysis: breaking down a polymer
HO 1 2 3 4 H
Hydrolysis adds a water molecule, breaking a bond.
H_2O
HO 1 2 3 H
HO H

The Diversity of Polymers

a) Each cell has thousands of different macromolecules

b) Macromolecules vary among cells of an organism, vary more within a species, and vary even more between species

c) A huge variety of polymers can be built from a small set of monomers

Concept 5.2: Carbohydrates serve as fuel and building material

a) **Carbohydrates** include sugars and the polymers of sugars

b) The simplest carbohydrates are monosaccharides, or simple sugars

c) Carbohydrate macromolecules are polysaccharides, polymers composed of many sugar building blocks

Sugars

a) **Monosaccharides** have molecular formulas that are usually multiples of CH_2O

b) Glucose ($C_6H_{12}O_6$) is the most common monosaccharide

c) Monosaccharides are classified by

   a) The location of the carbonyl group (as aldose or ketose)

   b) The number of carbons in the carbon skeleton

Figure 5.3

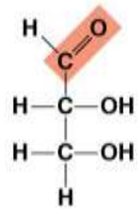
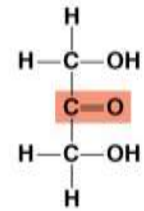
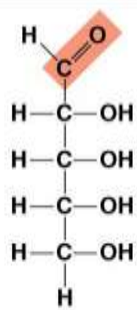
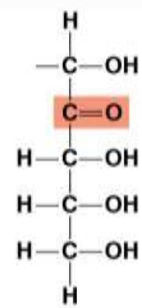
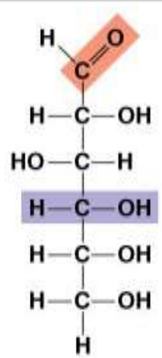
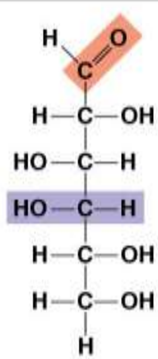
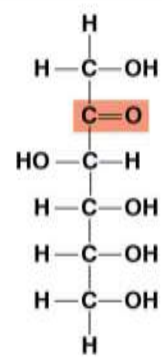
Aldoses (Aldehyde Sugars)	Ketoses (Ketone Sugars)
Trioses: 3-carbon sugars ($C_3H_6O_3$)	
 Glyceraldehyde	 Dihydroxyacetone
Pentoses: 5-carbon sugars ($C_5H_{10}O_5$)	
 Ribose	 Ribulose
Hexoses: 6-carbon sugars ($C_6H_{12}O_6$)	
 Glucose  Galactose	 Fructose

Figure 5.3a

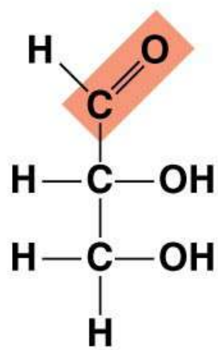
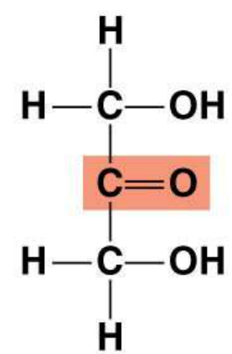
Aldose (Aldehyde Sugar)	Ketose (Ketone Sugar)
Trioses: 3-carbon sugars ($C_3H_6O_3$)	
 Glyceraldehyde	 Dihydroxyacetone

Figure 5.3b

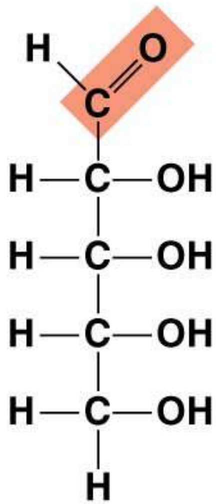
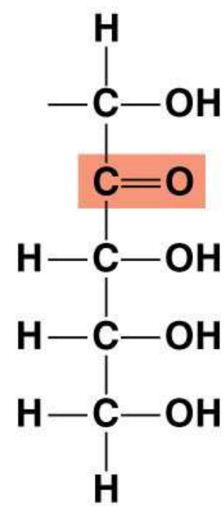
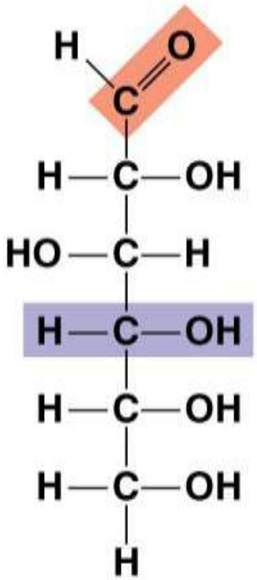
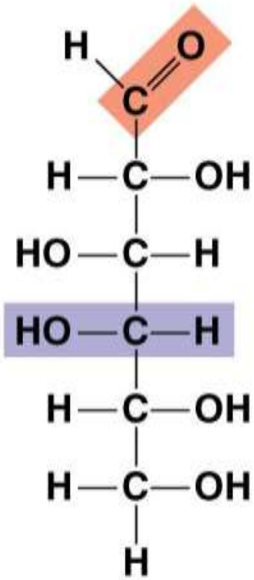
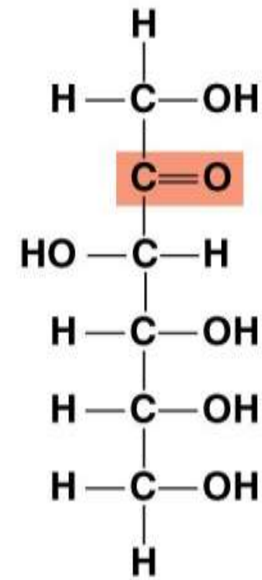
Aldose (Aldehyde Sugar)	Ketose (Ketone Sugar)
Pentoses: 5-carbon sugars ($C_5H_{10}O_5$)	
 Ribose	 Ribulose

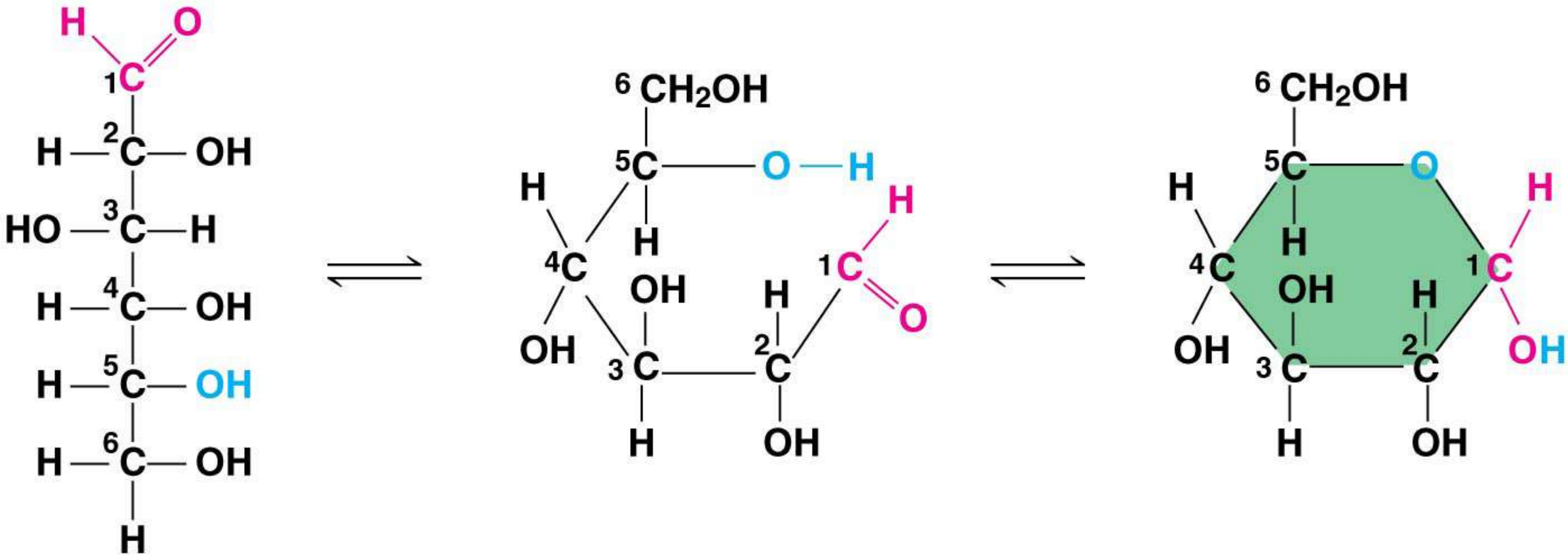
Figure 5.3c

<table>
<tr><th colspan="2">Aldose (Aldehyde Sugar)</th><th>Ketose (Ketone Sugar)</th></tr>
<tr><td colspan="3">Hexoses: 6-carbon sugars ($C_6H_{12}O_6$)</td></tr>
<tr><td>
Glucose</td><td>
Galactose</td><td>
Fructose</td></tr>
</table>

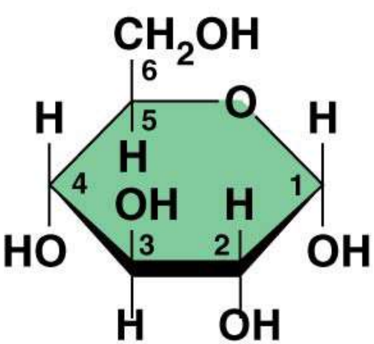
a) Though often drawn as linear skeletons, in aqueous solutions many sugars form rings

b) Monosaccharides serve as a major fuel for cells and as raw material for building molecules

Figure 5.4



(a) Linear and ring forms



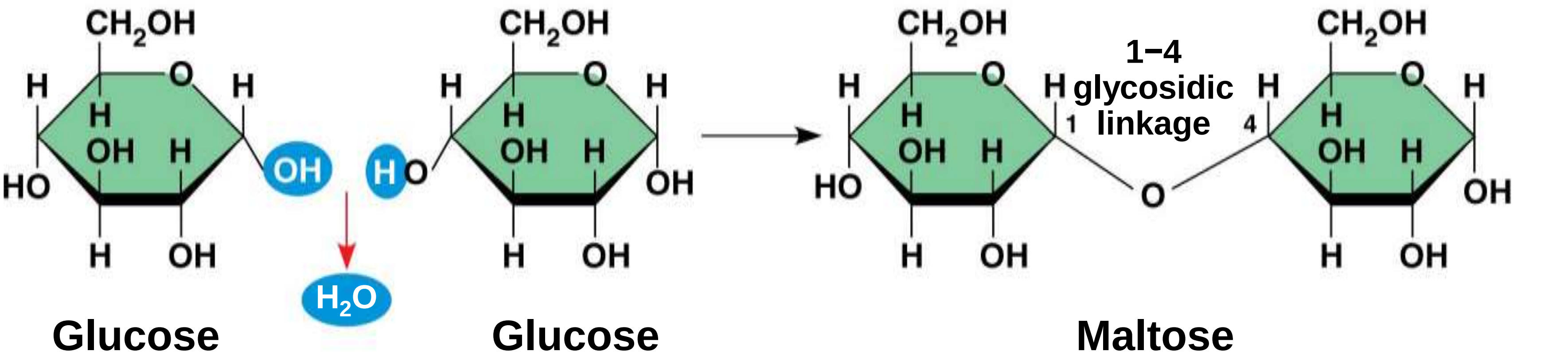
(b) Abbreviated ring structure

a) A **disaccharide** is formed when a dehydration reaction joins two monosaccharides

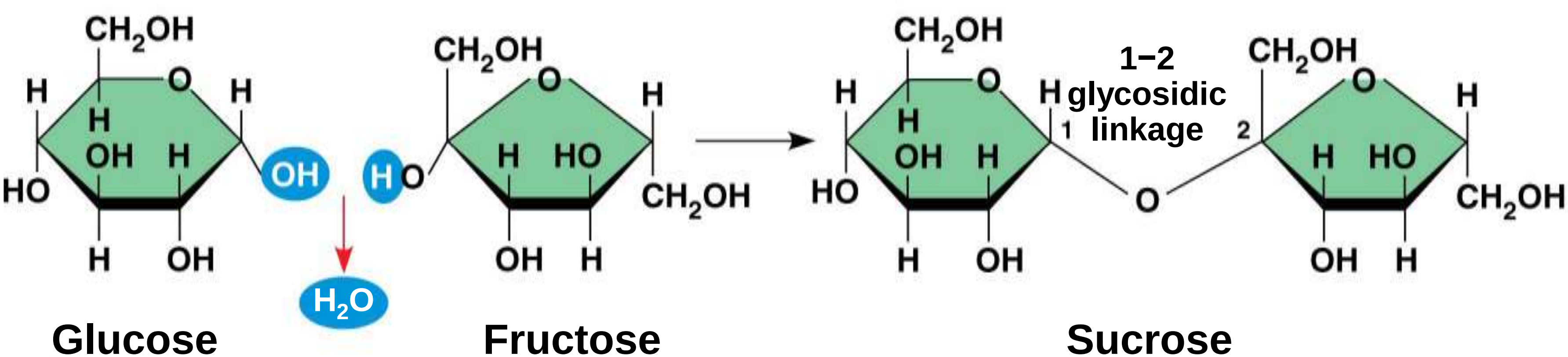
b) This covalent bond is called a **glycosidic linkage**

Figure 5.5

(a) Dehydration reaction in the synthesis of maltose



(b) Dehydration reaction in the synthesis of sucrose



Polysaccharides

a) **Polysaccharides**, the polymers of sugars, have storage and structural roles

b) The architecture and function of a polysaccharide are determined by its sugar monomers and the positions of its glycosidic linkages

Storage Polysaccharides

a) **Starch**, a storage polysaccharide of plants, consists entirely of glucose monomers

b) Plants store surplus starch as granules within chloroplasts and other plastids

c) The simplest form of starch is amylose

Figure 5.6

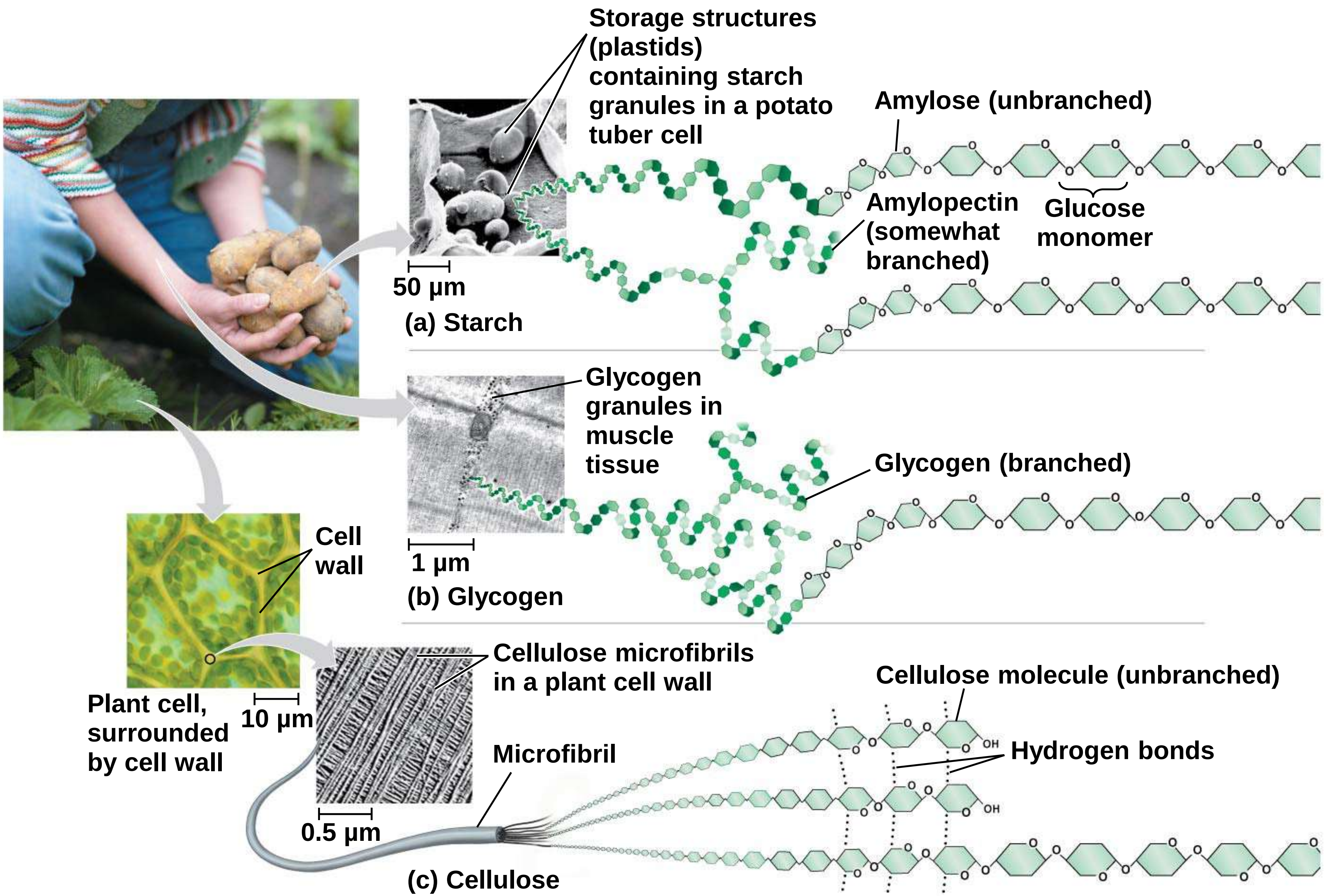
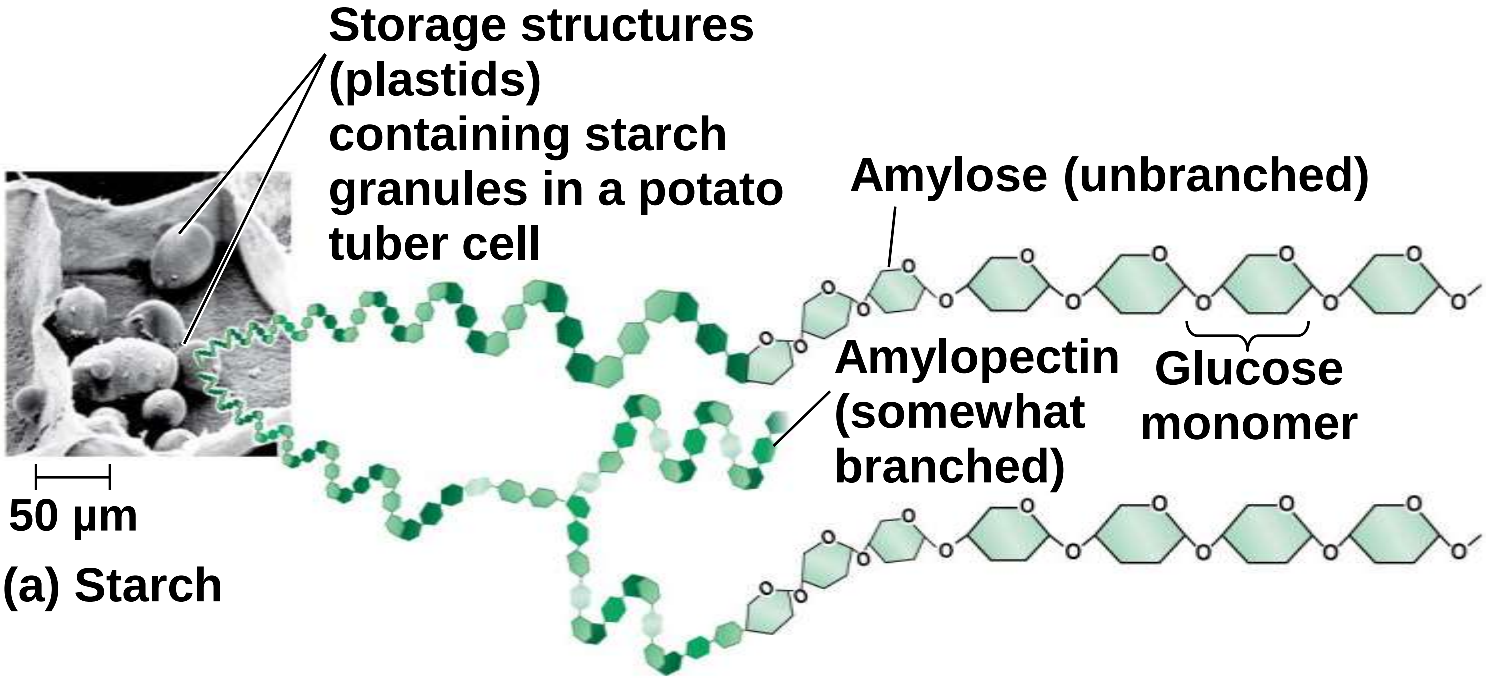

Storage structures (plastids) containing starch granules in a potato tuber cell
Amylose (unbranched)
Amylopectin (somewhat branched)
Glucose monomer
50 µm
(a) Starch
Glycogen granules in muscle tissue
Glycogen (branched)
1 µm
(b) Glycogen
Cell wall
Plant cell, surrounded by cell wall
10 µm
Cellulose microfibrils in a plant cell wall
Cellulose molecule (unbranched)
Hydrogen bonds
Microfibril
0.5 µm
(c) Cellulose

Figure 5.6a



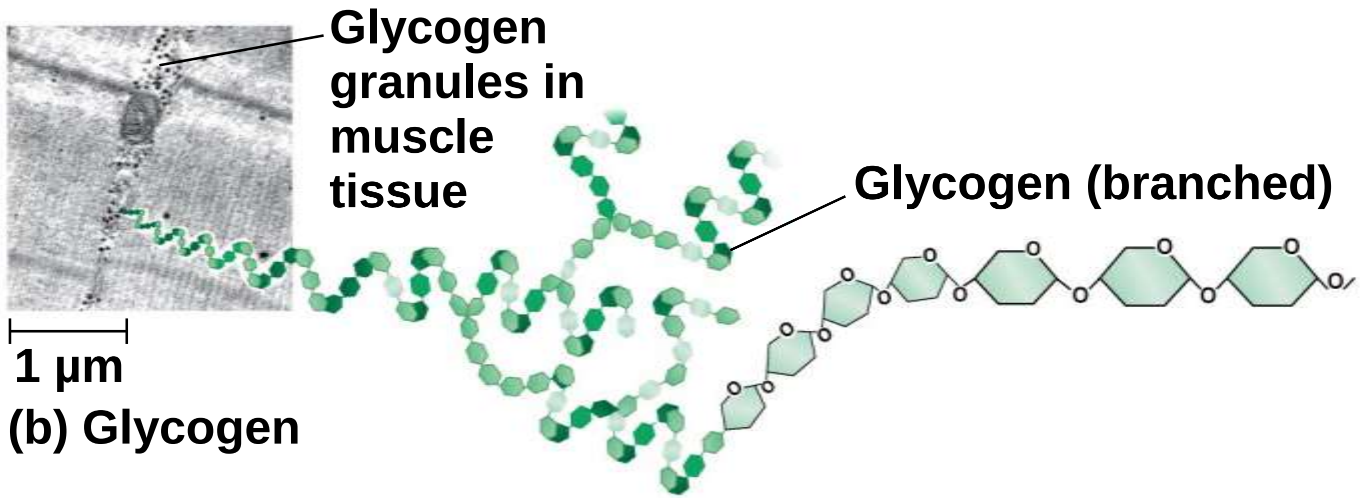
(a) Starch

a) **Glycogen** is a storage polysaccharide in animals

b) Glycogen is stored mainly in liver and muscle cells

c) Hydrolysis of glycogen in these cells releases glucose when the demand for sugar increases

Figure 5.6b



(b) Glycogen

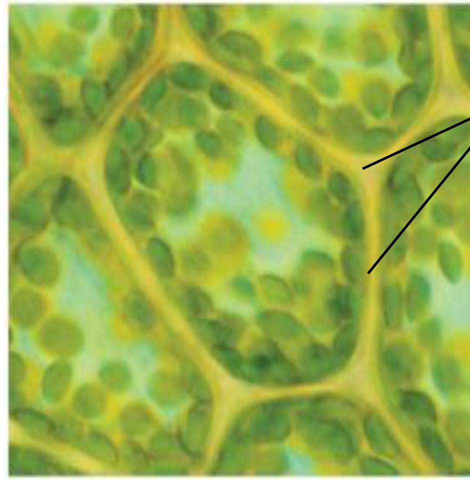
Structural Polysaccharides

a) The polysaccharide **cellulose** is a major component of the tough wall of plant cells

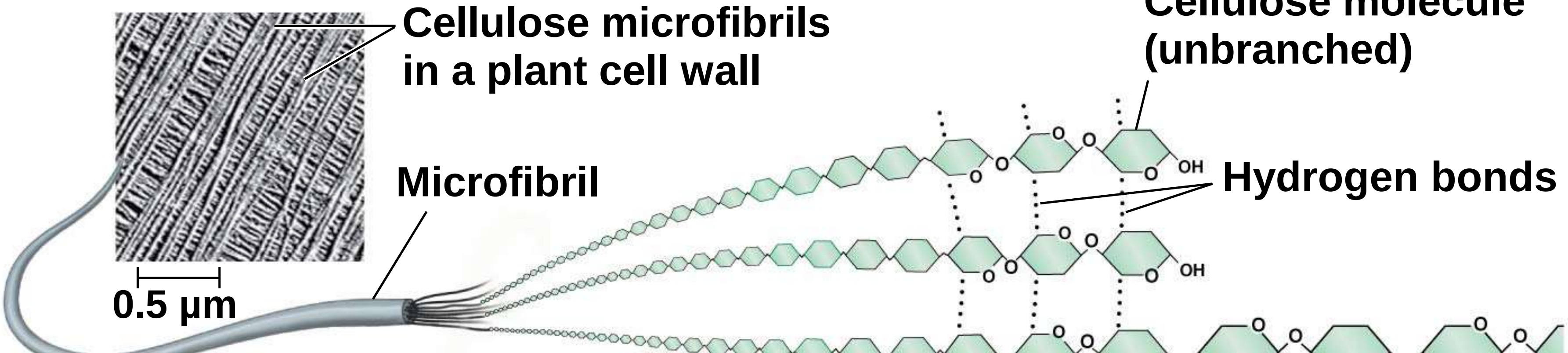
b) Like starch, cellulose is a polymer of glucose, but the glycosidic linkages differ

c) The difference is based on two ring forms for glucose: alpha (α) and beta (β)

Figure 5.6c

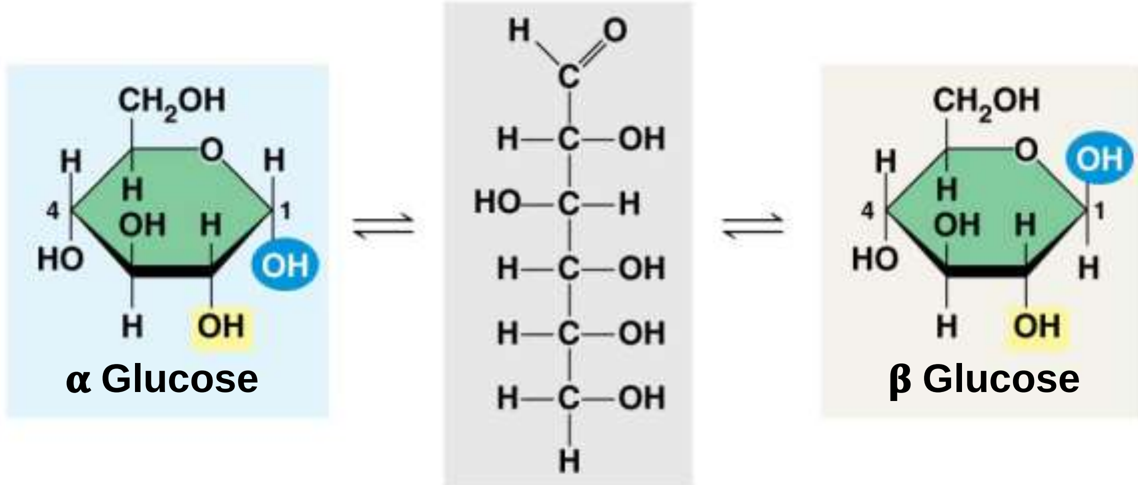


Plant cell, surrounded by cell wall

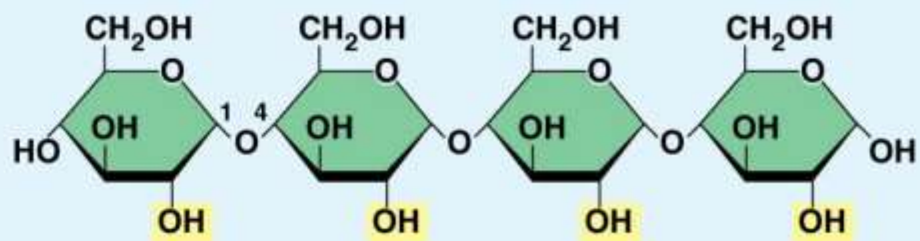


(c) Cellulose

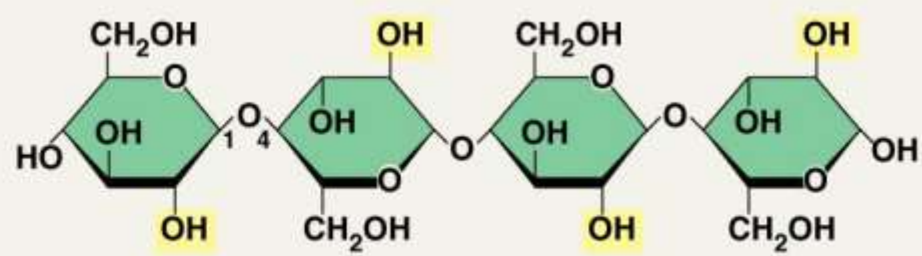
Figure 5.7



(a) α and β glucose ring structures

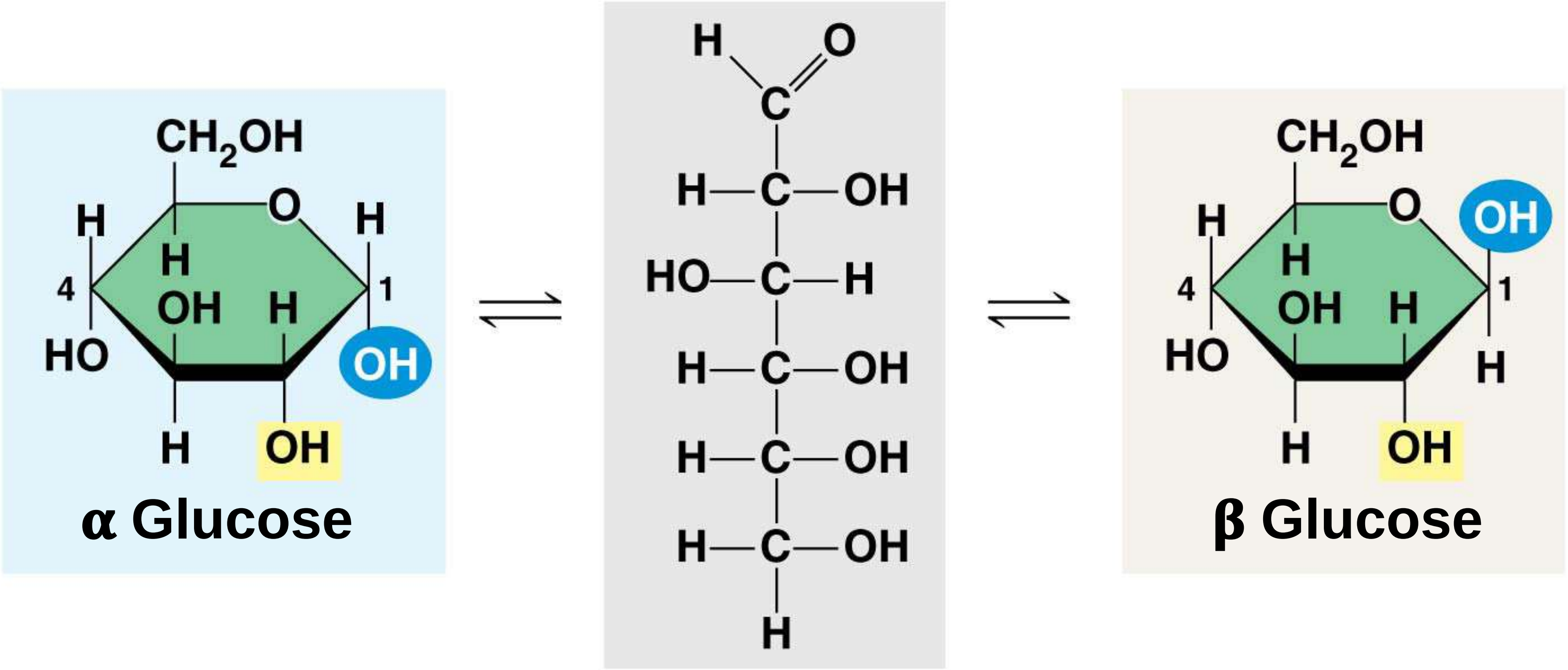


(b) Starch: 1–4 linkage of α glucose monomer



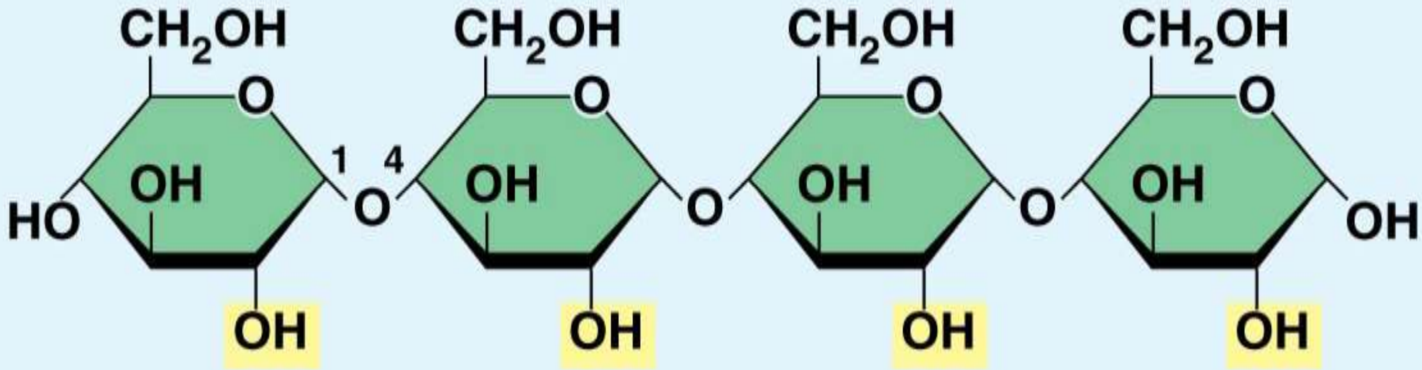
(c) Cellulose: 1–4 linkage of β glucose monomers

Figure 5.7a

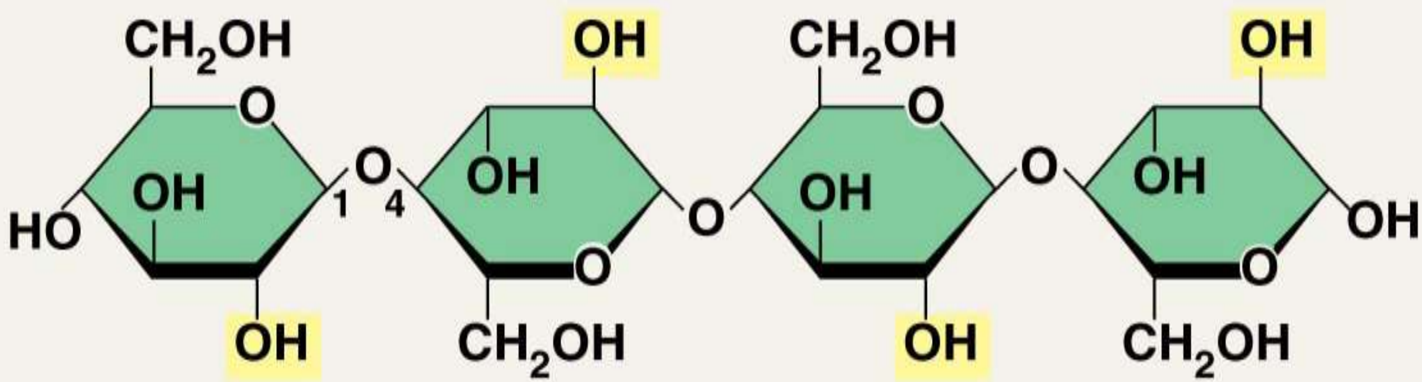


(a) α and β glucose ring structures

Figure 5.7b



(b) Starch: 1–4 linkage of α glucose monomer



(c) Cellulose: 1–4 linkage of β glucose monomers

a) Starch (α configuration) is largely helical

b) Cellulose molecules (β configuration) are straight and unbranched

c) Some hydroxyl groups on the monomers of cellulose can hydrogen bond with hydroxyls of parallel cellulose molecules

a) Enzymes that digest starch by hydrolyzing α linkages can't hydrolyze β linkages in cellulose

b) The cellulose in human food passes through the digestive tract as "insoluble fiber"

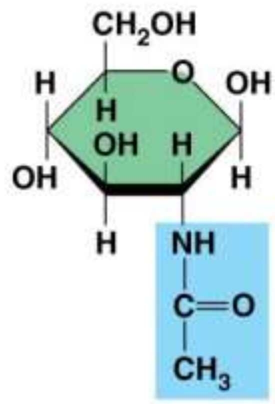
c) Some microbes use enzymes to digest cellulose

d) Many herbivores, from cows to termites, have symbiotic relationships with these microbes

a) **Chitin**, another structural polysaccharide, is found in the exoskeleton of arthropods

b) Chitin also provides structural support for the cell walls of many fungi

Figure 5.8



The structure of the chitin monomer

Chitin, embedded in proteins, forms the exoskeleton of arthropods.

Chitin is used to make a strong and flexible surgical thread.

Concept 5.3: Lipids are a diverse group of hydrophobic molecules

a) **Lipids** are the one class of large biological molecules that does not include true polymers

b) The unifying feature of lipids is that they mix poorly, if at all, with water

c) Lipids are hydrophobic because they consist mostly of hydrocarbons, which form nonpolar covalent bonds

d) The most biologically important lipids are fats, phospholipids, and steroids

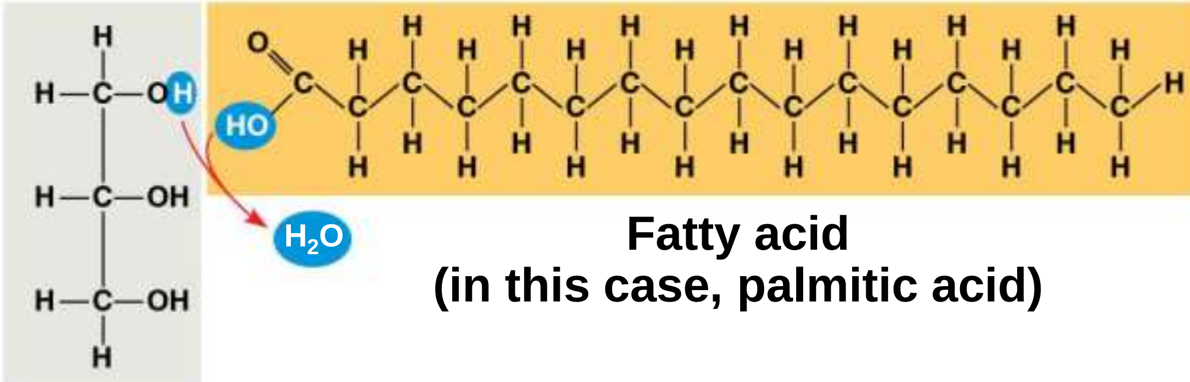
Fats

a) **Fats** are constructed from two types of smaller molecules: glycerol and fatty acids

b) Glycerol is a three-carbon alcohol with a hydroxyl group attached to each carbon

c) A **fatty acid** consists of a carboxyl group attached to a long carbon skeleton

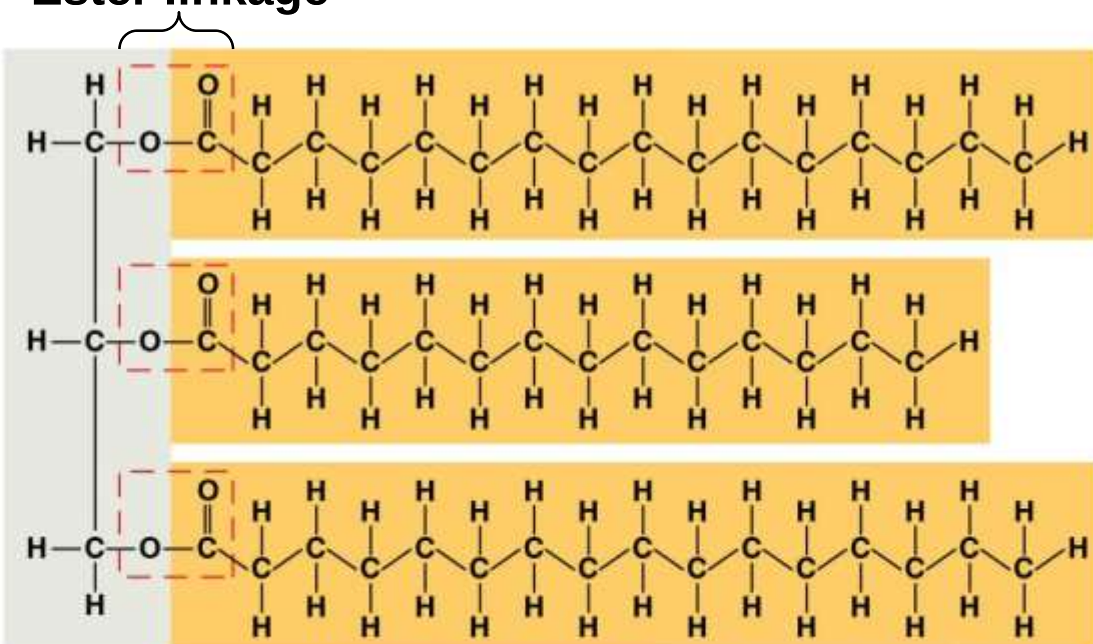
Figure 5.9



Glycerol

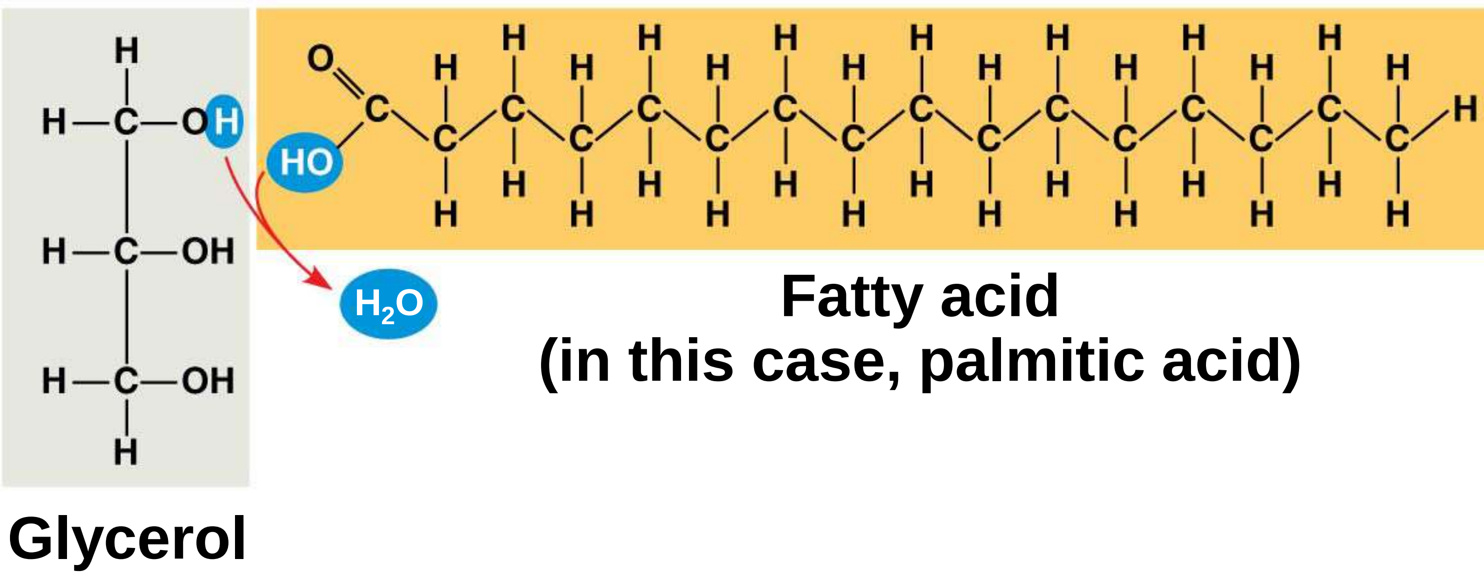
(a) One of three dehydration reactions in the synthesis of a fat

Ester linkage



(b) Fat molecule (triacylglycerol)

Figure 5.9a



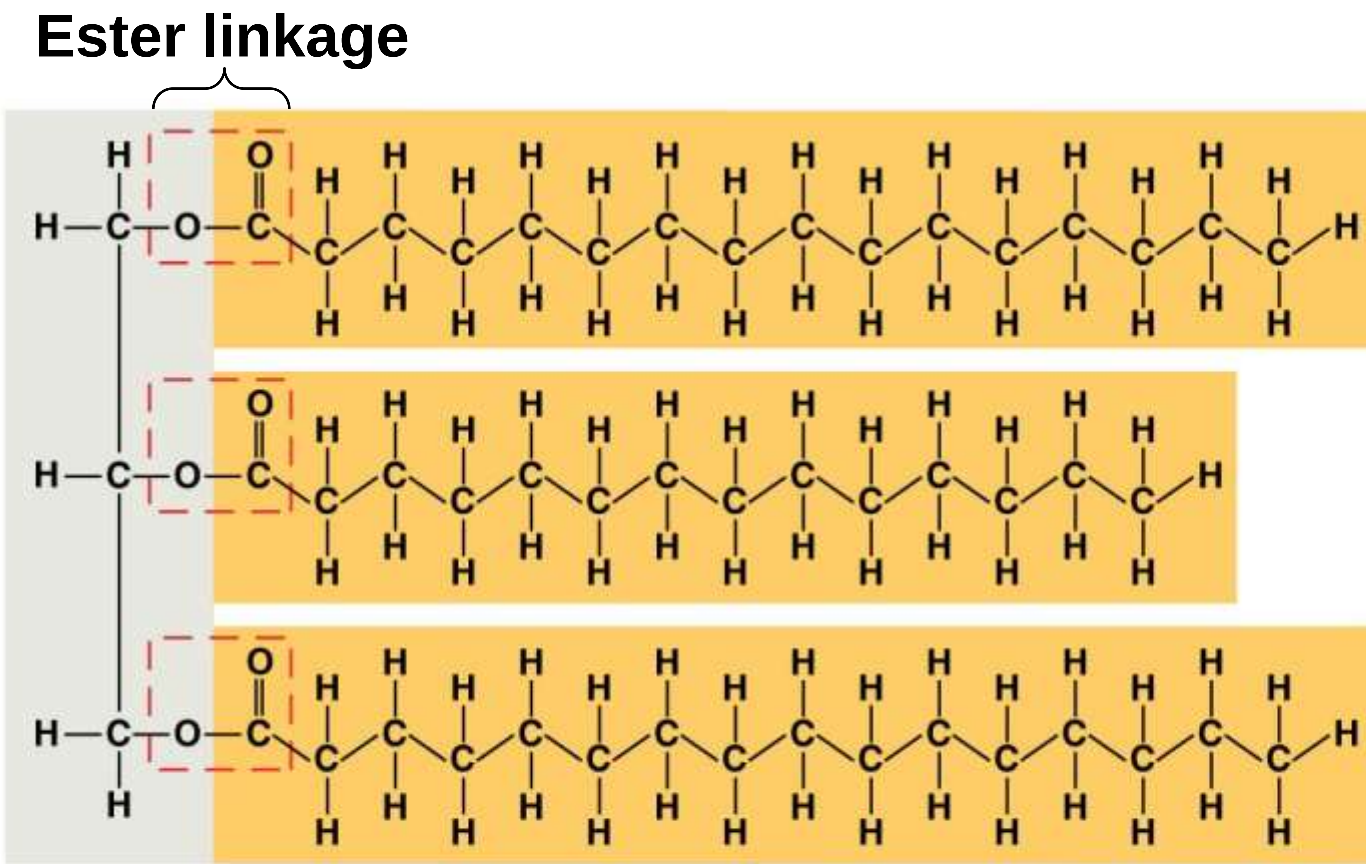
(a) One of three dehydration reactions in the synthesis of a fat

a) Fats separate from water because water molecules hydrogen-bond to each other and exclude the fats

b) In a fat, three fatty acids are joined to glycerol by an ester linkage, creating a **triacylglycerol**, or triglyceride

c) The fatty acids in a fat can be all the same or of two or three different kinds

Figure 5.9b



(b) Fat molecule (triacylglycerol)

a) Fatty acids vary in length (number of carbons) and in the number and locations of double bonds

b) **Saturated fatty acids** have the maximum number of hydrogen atoms possible and no double bonds

c) **Unsaturated fatty acids** have one or more double bonds

Figure 5.10

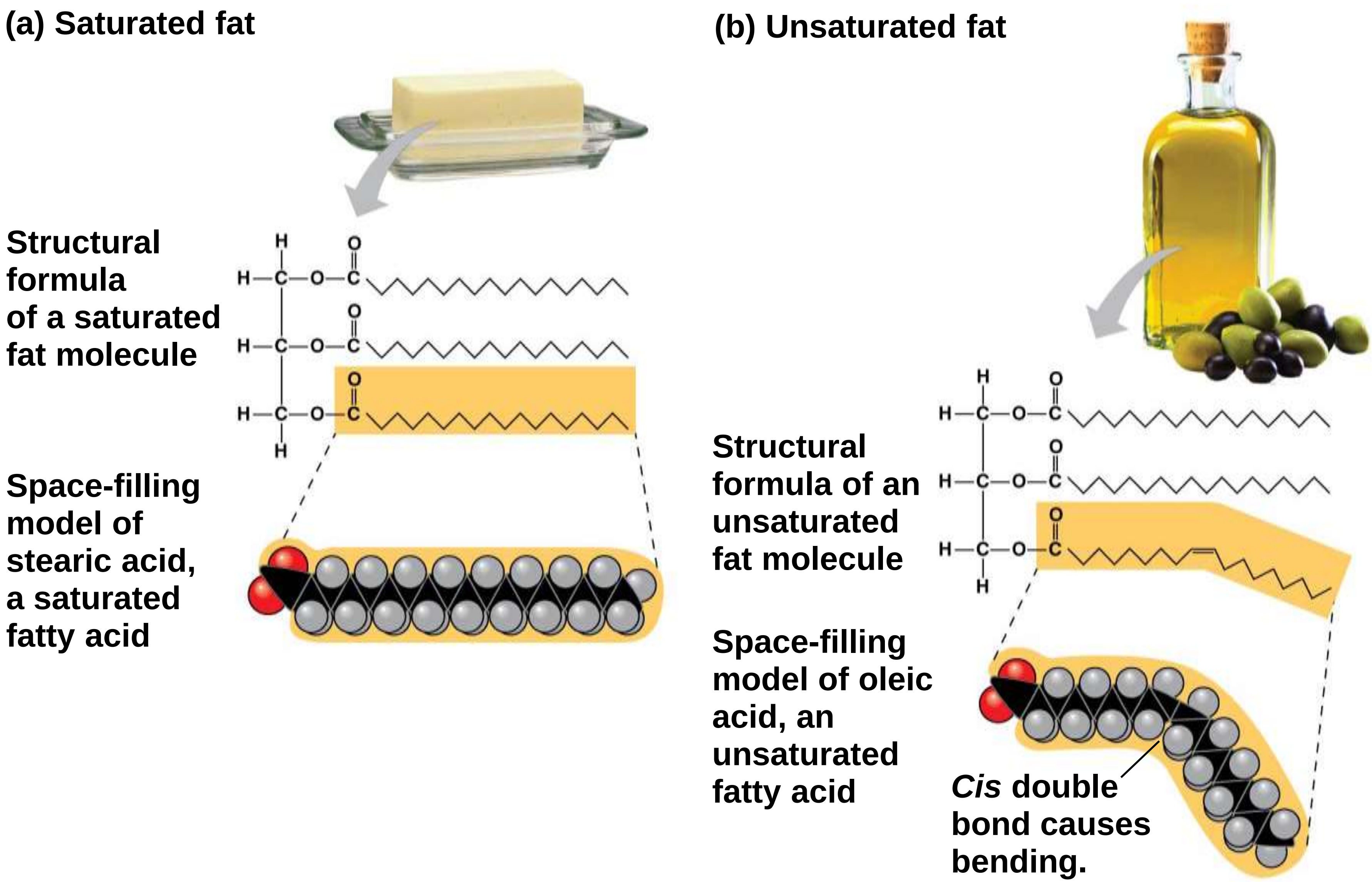
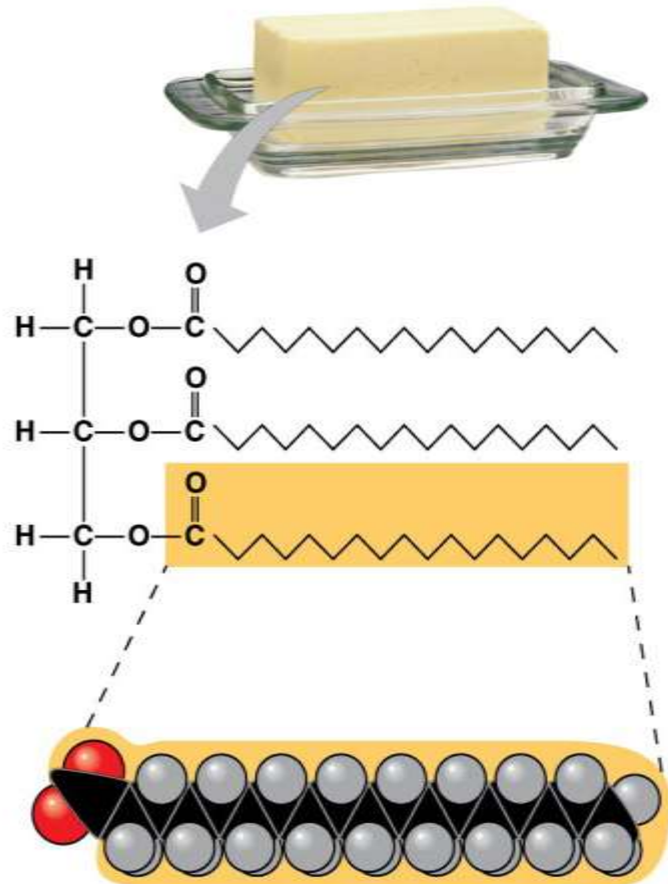

(a) Saturated fat
Structural formula of a saturated fat molecule
Space-filling model of stearic acid, a saturated fatty acid
(b) Unsaturated fat
Structural formula of an unsaturated fat molecule
Space-filling model of oleic acid, an unsaturated fatty acid
Cis double bond causes bending.

Figure 5.10a

(a) Saturated fat

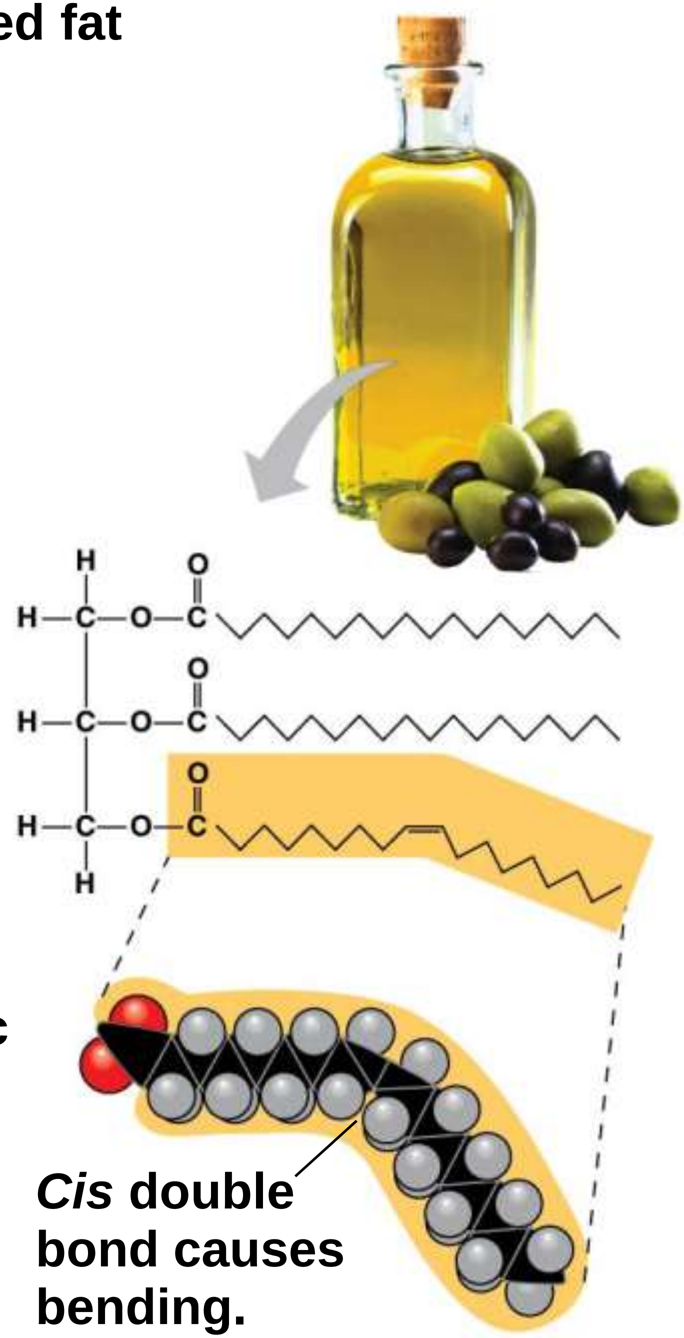


Structural formula of a saturated fat molecule

Space-filling model of stearic acid, a saturated fatty acid

Figure 5.10b

(b) Unsaturated fat



Structural formula of an unsaturated fat molecule

Space-filling model of oleic acid, an unsaturated fatty acid

a) Fats made from saturated fatty acids are called saturated fats and are solid at room temperature

b) Most animal fats are saturated

c) Fats made from unsaturated fatty acids are called unsaturated fats or oils and are liquid at room temperature

d) Plant fats and fish fats are usually unsaturated

a) A diet rich in saturated fats may contribute to cardiovascular disease through plaque deposits

b) Hydrogenation is the process of converting unsaturated fats to saturated fats by adding hydrogen

c) Hydrogenating vegetable oils also creates unsaturated fats with *trans* double bonds

d) These ***trans* fats** may contribute more than saturated fats to cardiovascular disease

a) Certain unsaturated fatty acids are not synthesized in the human body

b) These must be supplied in the diet

c) These essential fatty acids include the omega-3 fatty acids, which are required for normal growth and are thought to provide protection against cardiovascular disease

a) The major function of fats is energy storage

b) Humans and other mammals store their long-term food reserves in adipose cells

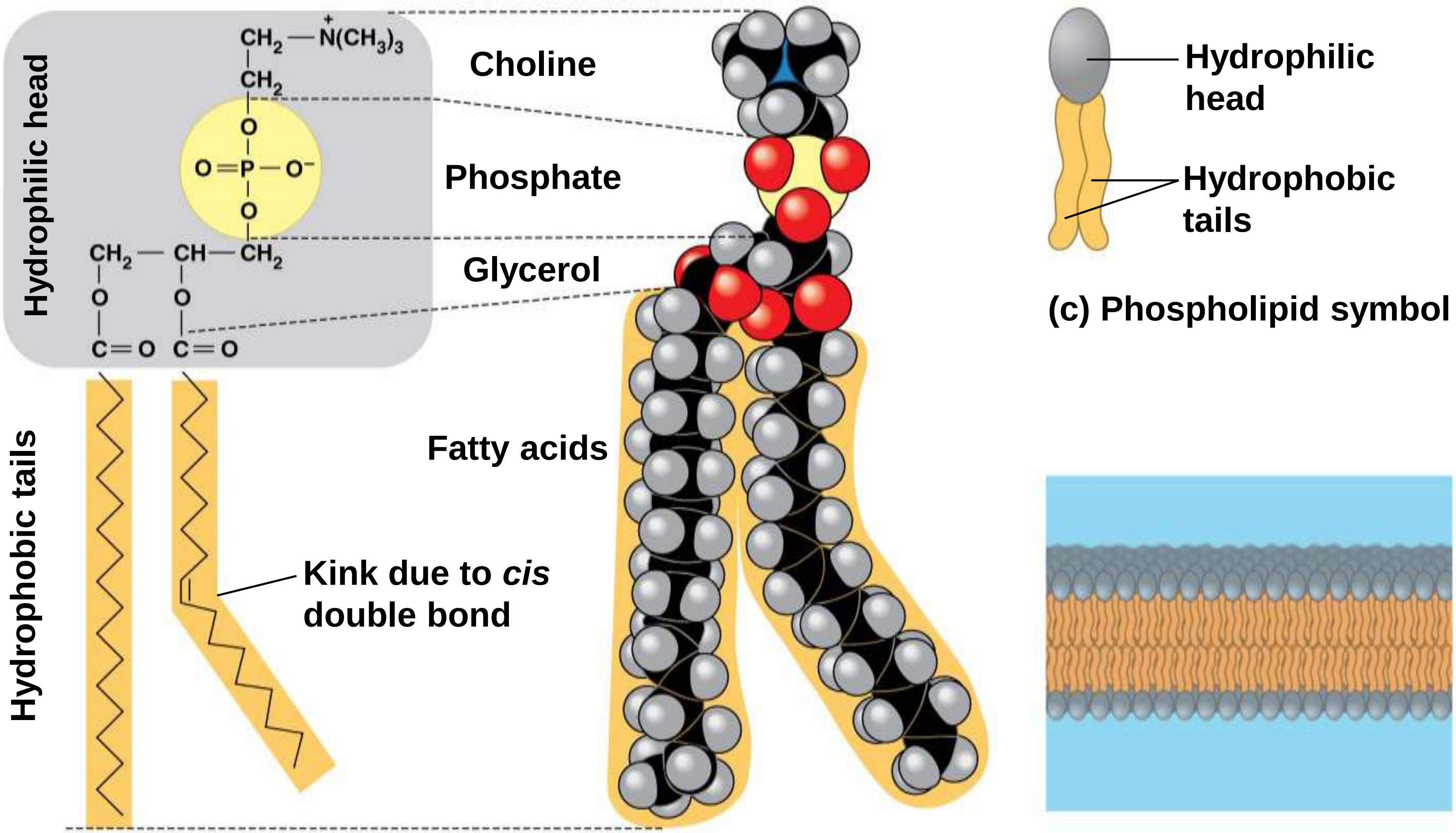
c) Adipose tissue also cushions vital organs and insulates the body

Phospholipids

a) In a **phospholipid**, two fatty acids and a phosphate group are attached to glycerol

b) The two fatty acid tails are hydrophobic, but the phosphate group and its attachments form a hydrophilic head

Figure 5.11



(a) Structural formula

(b) Space-filling model

(c) Phospholipid symbol

(d) Phospholipid bilayer

a) When phospholipids are added to water, they self-assemble into double-layered structures called bilayers

b) At the surface of a cell, phospholipids are also arranged in a bilayer, with the hydrophobic tails pointing toward the interior

c) The structure of phospholipids results in a bilayer arrangement found in cell membranes

d) The existence of cells depends on phospholipids

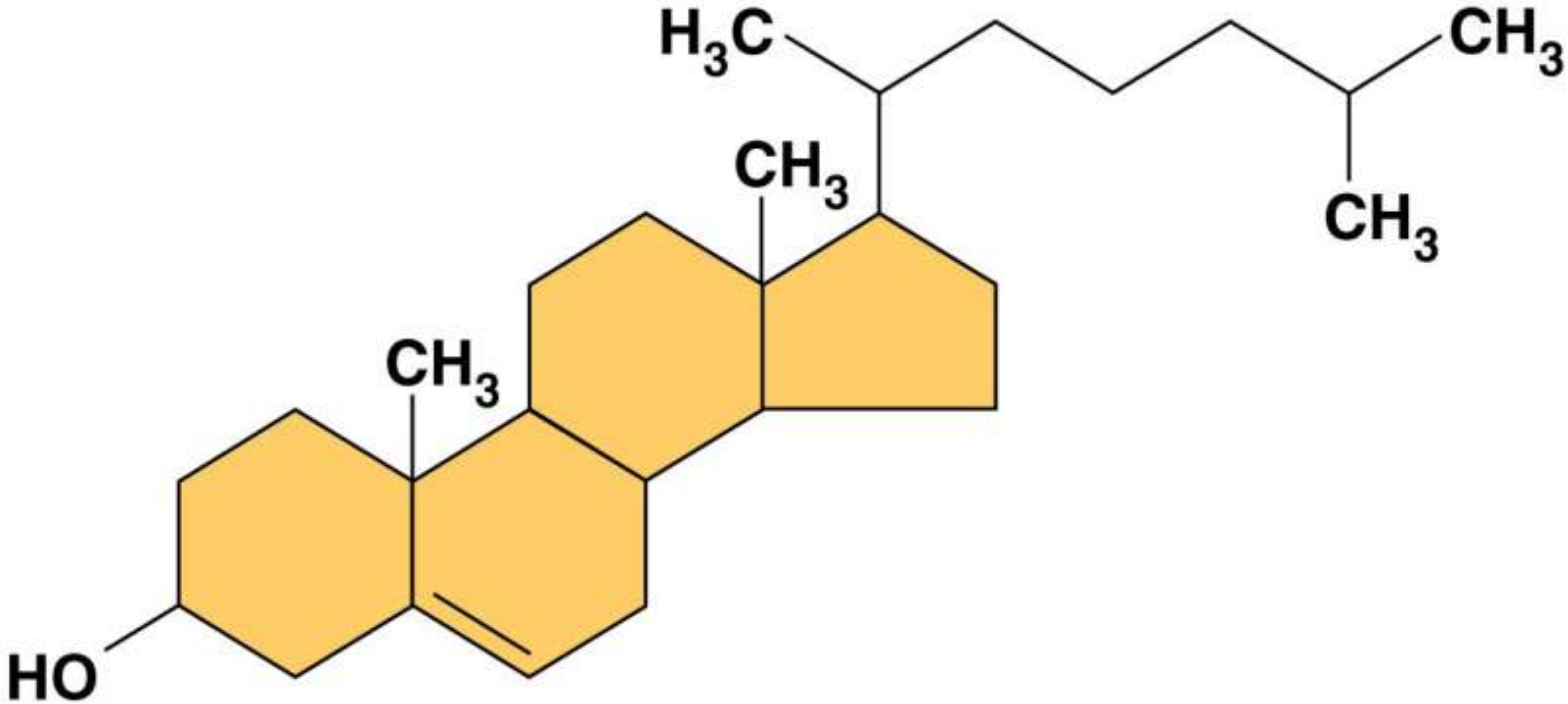
Steroids

a) **Steroids** are lipids characterized by a carbon skeleton consisting of four fused rings

b) **Cholesterol**, a type of steroid, is a component in animal cell membranes and a precursor from which other steroids are synthesized

c) A high level of cholesterol in the blood may contribute to cardiovascular disease

Figure 5.12



Concept 5.4: Proteins include a diversity of structures, resulting in a wide range of functions

a) Proteins account for more than 50% of the dry mass of most cells

b) Some proteins speed up chemical reactions

c) Other protein functions include defense, storage, transport, cellular communication, movement, or structural support

Figure 5.13a

Enzymatic proteins

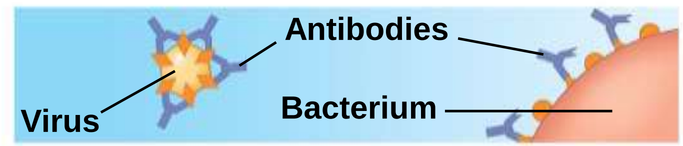
Function: Selective acceleration of chemical reactions

Example: Digestive enzymes catalyze the hydrolysis of bonds in food molecules.

Defensive proteins

Function: Protection against disease

Example: Antibodies inactivate and help destroy viruses and bacteria.



Storage proteins

Function: Storage of amino acids

Examples: Casein, the protein of milk, is the major source of amino acids for baby mammals. Plants have storage proteins in their seeds. Ovalbumin is the protein of egg white, used as an amino acid source for the developing embryo.

Transport proteins

Function: Transport of substances

Examples: Hemoglobin, the iron-containing protein of vertebrate blood, transports oxygen from the lungs to other parts of the body. Other proteins transport molecules across membranes, as shown here.

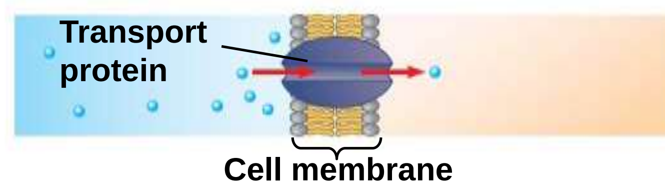
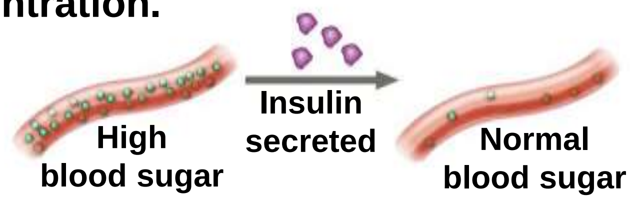


Figure 5.13b

Hormonal proteins

Function: Coordination of an organism's activities

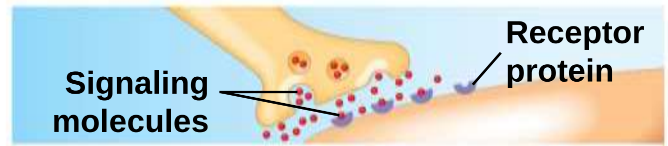
Example: Insulin, a hormone secreted by the pancreas, causes other tissues to take up glucose, thus regulating blood sugar, concentration.



Receptor proteins

Function: Response of cell to chemical stimuli

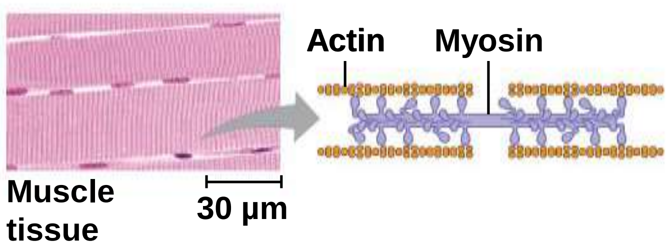
Example: Receptors built into the membrane of a nerve cell detect signaling molecules released by other nerve cells.



Contractile and motor proteins

Function: Movement

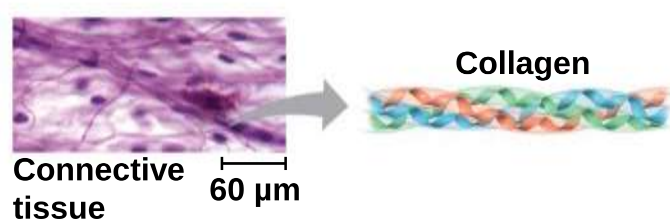
Examples: Motor proteins are responsible for the undulations of cilia and flagella. Actin and myosin proteins are responsible for the contraction of muscles.



Structural proteins

Function: Support

Examples: Keratin is the protein of hair, horns, feathers, and other skin appendages. Insects and spiders use silk fibers to make their cocoons and webs, respectively. Collagen and elastin proteins provide a fibrous framework in animal connective tissues.



a) Enzymes are proteins that act as **catalysts** to speed up chemical reactions

b) Enzymes can perform their functions repeatedly, functioning as workhorses that carry out the processes of life

a) Proteins are all constructed from the same set of 20 amino acids

b) Polypeptides are unbranched polymers built from these amino acids

c) A **protein** is a biologically functional molecule that consists of one or more polypeptides

Amino Acid Monomers

a) **Amino acids** are organic molecules with amino and carboxyl groups

b) Amino acids differ in their properties due to differing side chains, called R groups

Figure 5.UN01

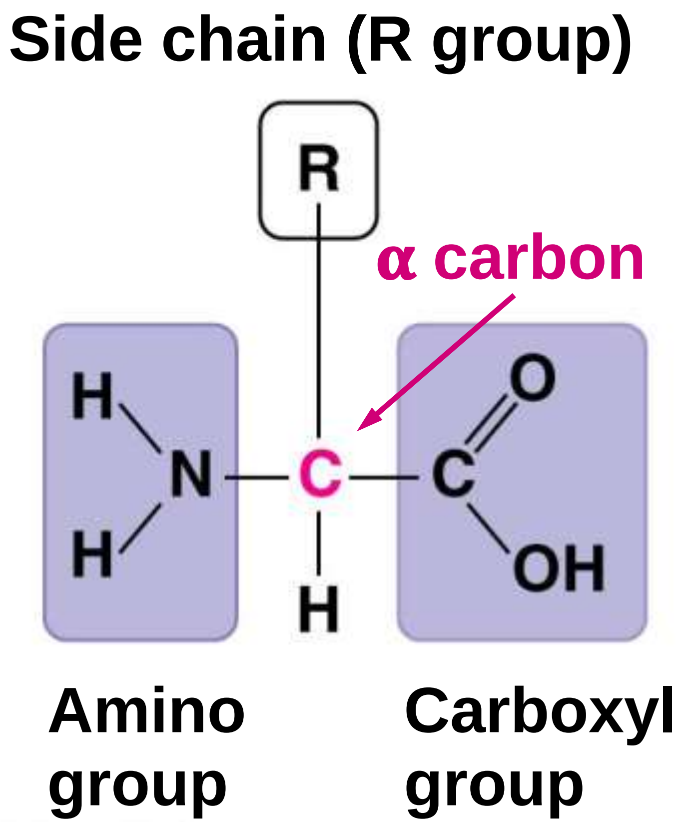
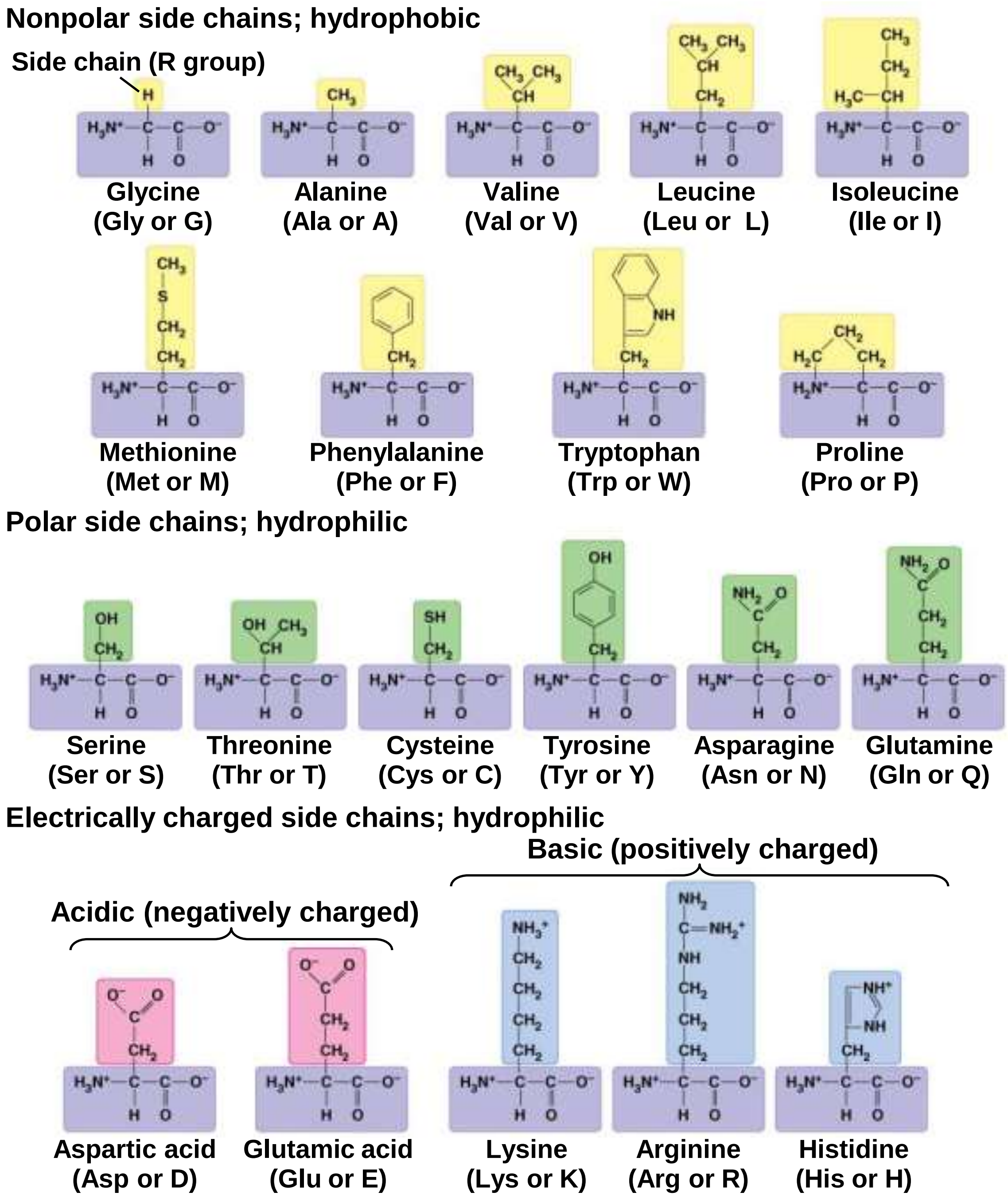


Figure 5.14


Nonpolar side chains; hydrophobic
Side chain (R group)
Glycine (Gly or G)
Alanine (Ala or A)
Valine (Val or V)
Leucine (Leu or L)
Isoleucine (Ile or I)
Methionine (Met or M)
Phenylalanine (Phe or F)
Tryptophan (Trp or W)
Proline (Pro or P)
Polar side chains; hydrophilic
Serine (Ser or S)
Threonine (Thr or T)
Cysteine (Cys or C)
Tyrosine (Tyr or Y)
Asparagine (Asn or N)
Glutamine (Gln or Q)
Electrically charged side chains; hydrophilic
Basic (positively charged)
Acidic (negatively charged)
Aspartic acid (Asp or D)
Glutamic acid (Glu or E)
Lysine (Lys or K)
Arginine (Arg or R)
Histidine (His or H)

Polypeptides (Amino Acid Polymers)

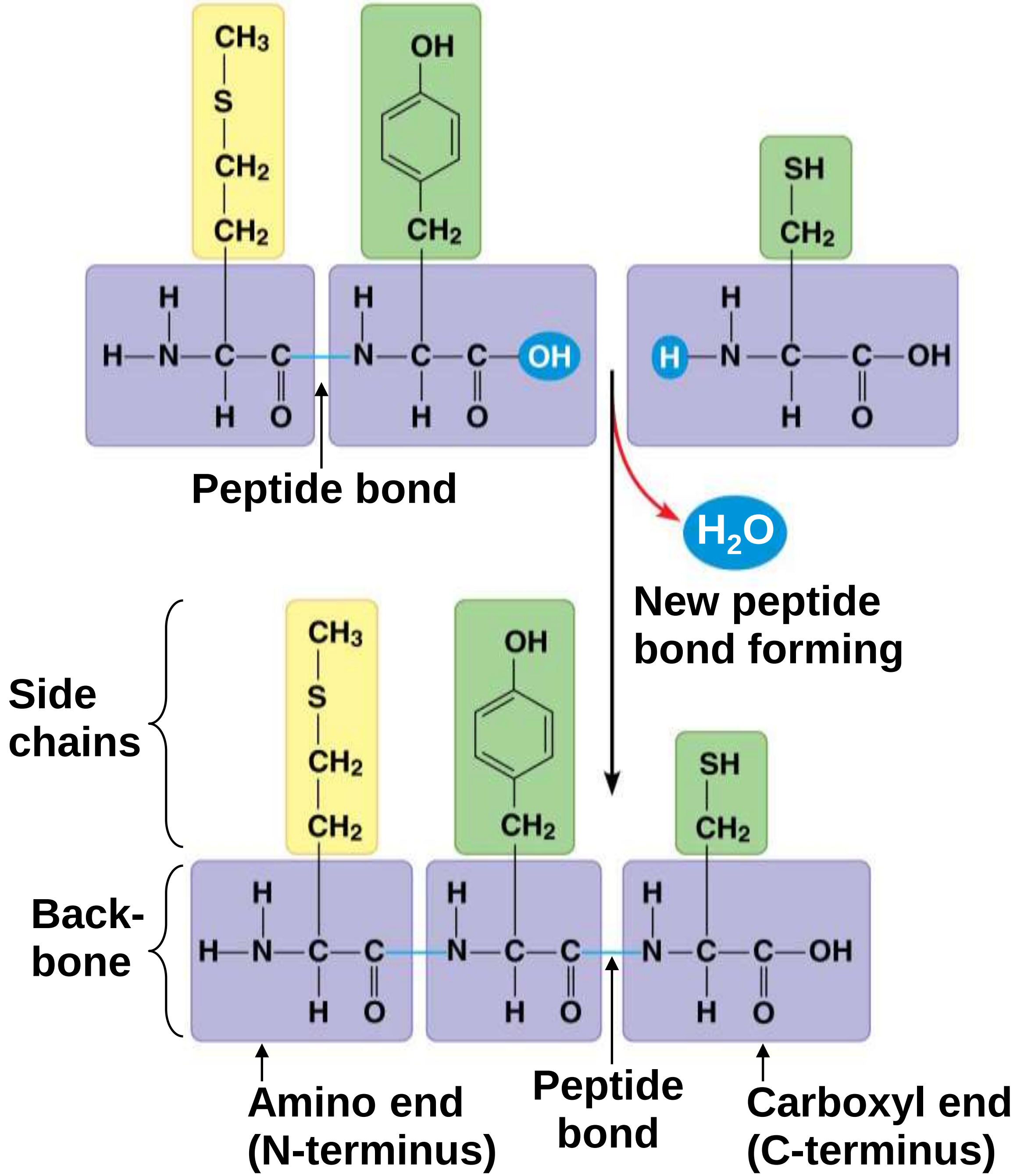
a) Amino acids are linked by covalent bonds called **peptide bonds**

b) A polypeptide is a polymer of amino acids

c) Polypeptides range in length from a few to more than a thousand monomers

d) Each polypeptide has a unique linear sequence of amino acids, with a carboxyl end (C-terminus) and an amino end (N-terminus)

Figure 5.15

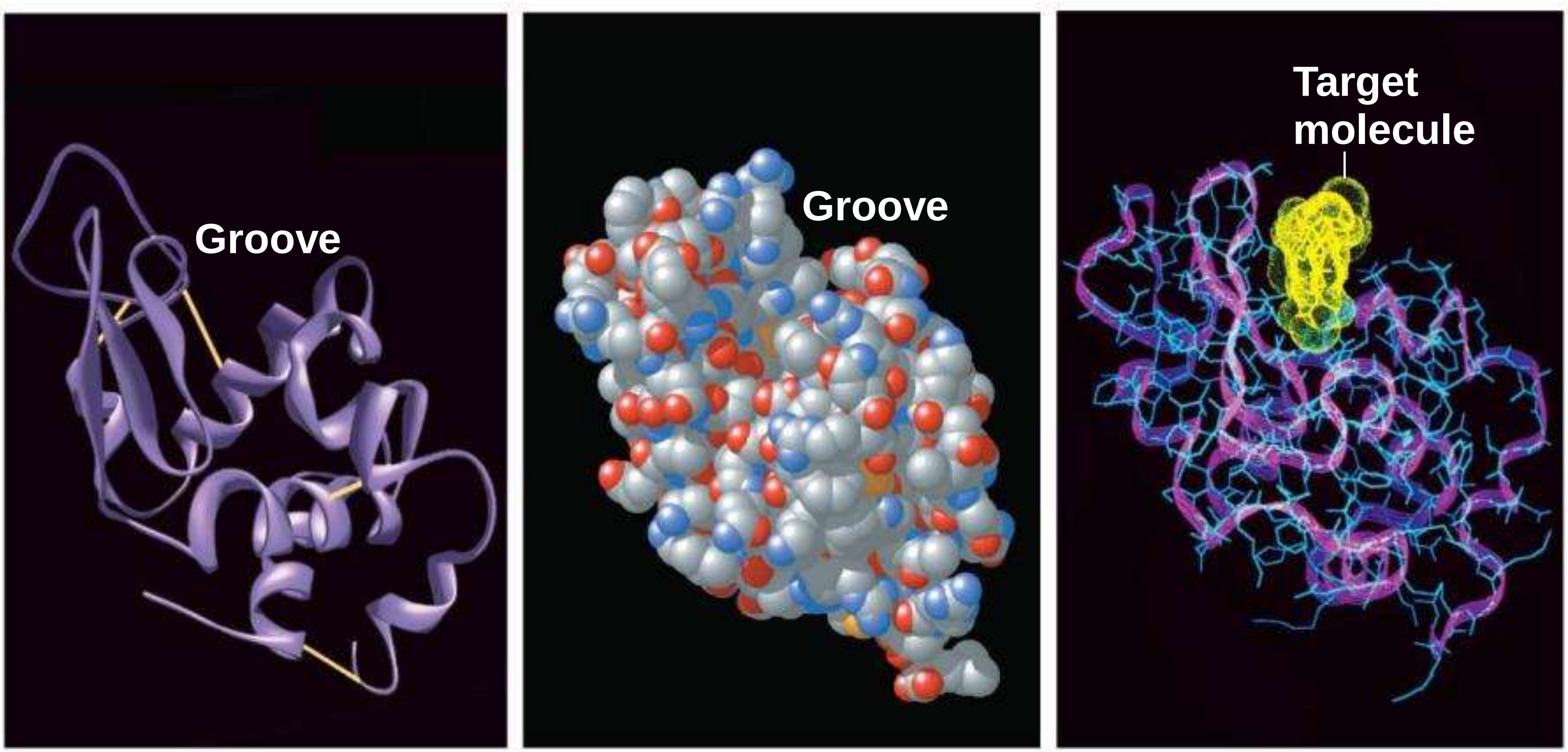

CH3
S
CH2
CH2
OH
CH2
SH
CH2
H
H—N—C—C
H O
H
N—C—C
H O
OH
H
H
N—C—C—OH
H O
Peptide bond
H_2O
New peptide
bond forming
Side
chains
Back-
bone
Amino end
(N-terminus)
Peptide
bond
Carboxyl end
(C-terminus)

Protein Structure and Function

a) The specific activities of proteins result from their intricate three-dimensional architecture

b) A functional protein consists of one or more polypeptides precisely twisted, folded, and coiled into a unique shape

Figure 5.16



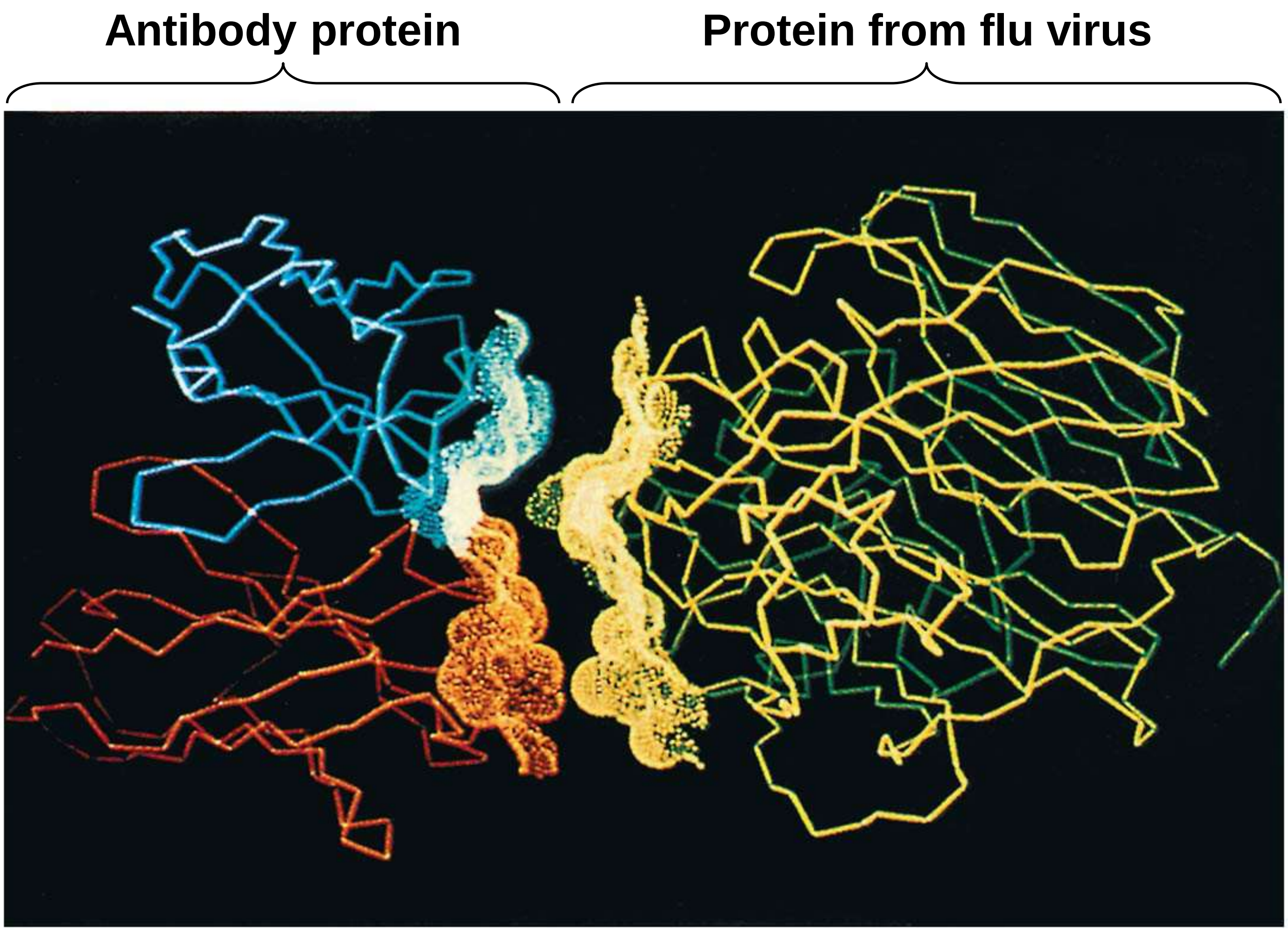
(a) A ribbon model (b) A space-filling model (c) A wireframe model

a) The sequence of amino acids determines a protein’s three-dimensional structure

b) A protein’s structure determines how it works

c) The function of a protein usually depends on its ability to recognize and bind to some other molecule

Figure 5.17



Four Levels of Protein Structure

a) The primary structure of a protein is its unique sequence of amino acids

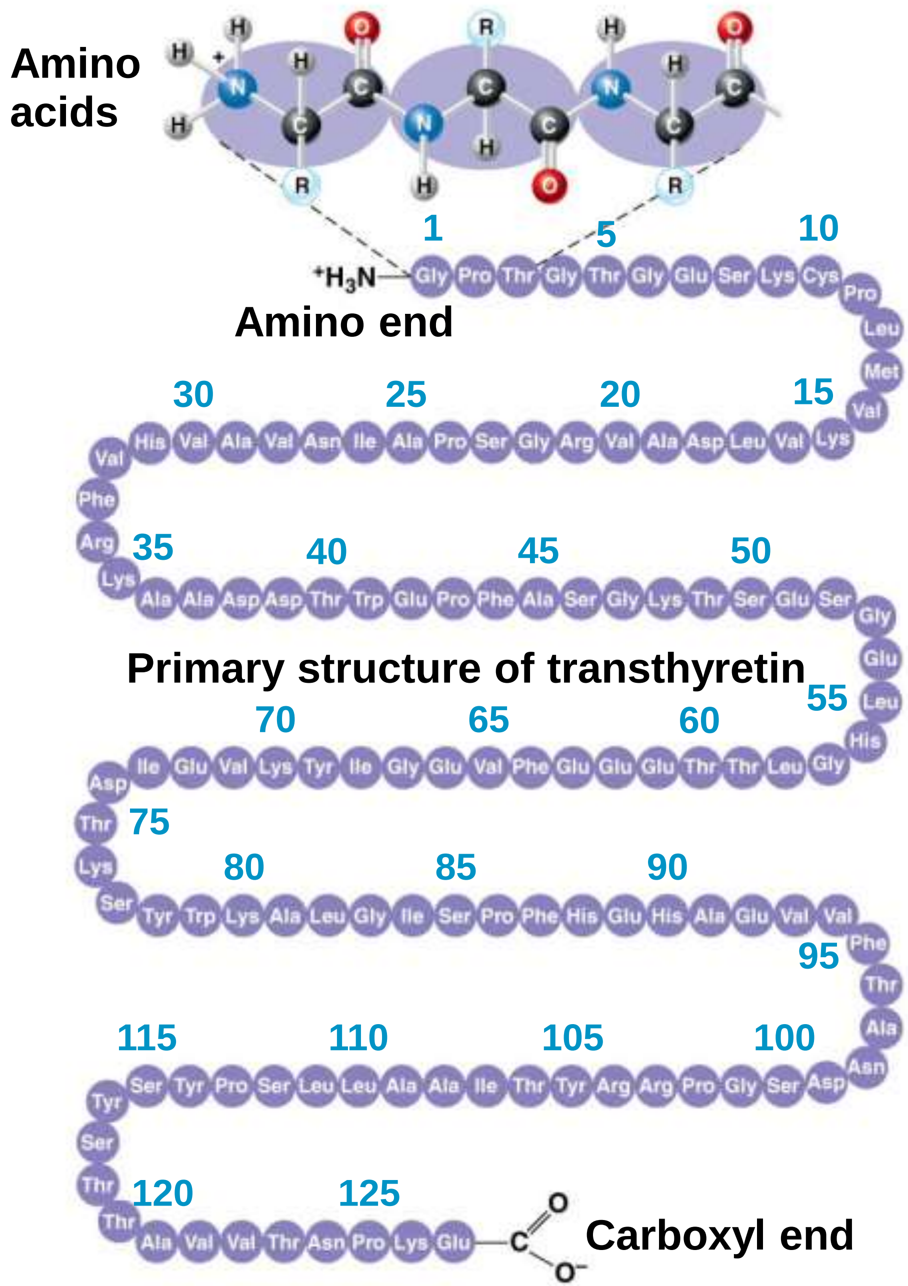
b) Secondary structure, found in most proteins, consists of coils and folds in the polypeptide chain

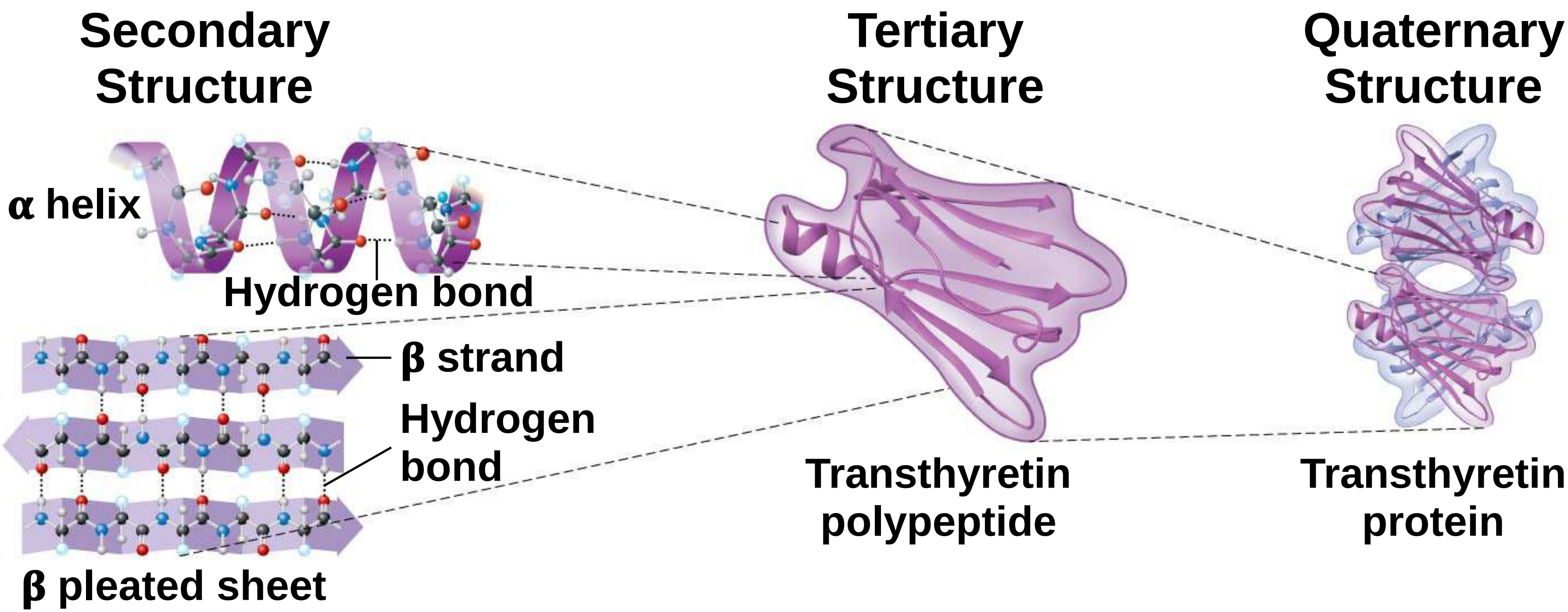
c) Tertiary structure is determined by interactions among various side chains (R groups)

d) Quaternary structure results when a protein consists of multiple polypeptide chains

Figure 5.18a

Primary Structure



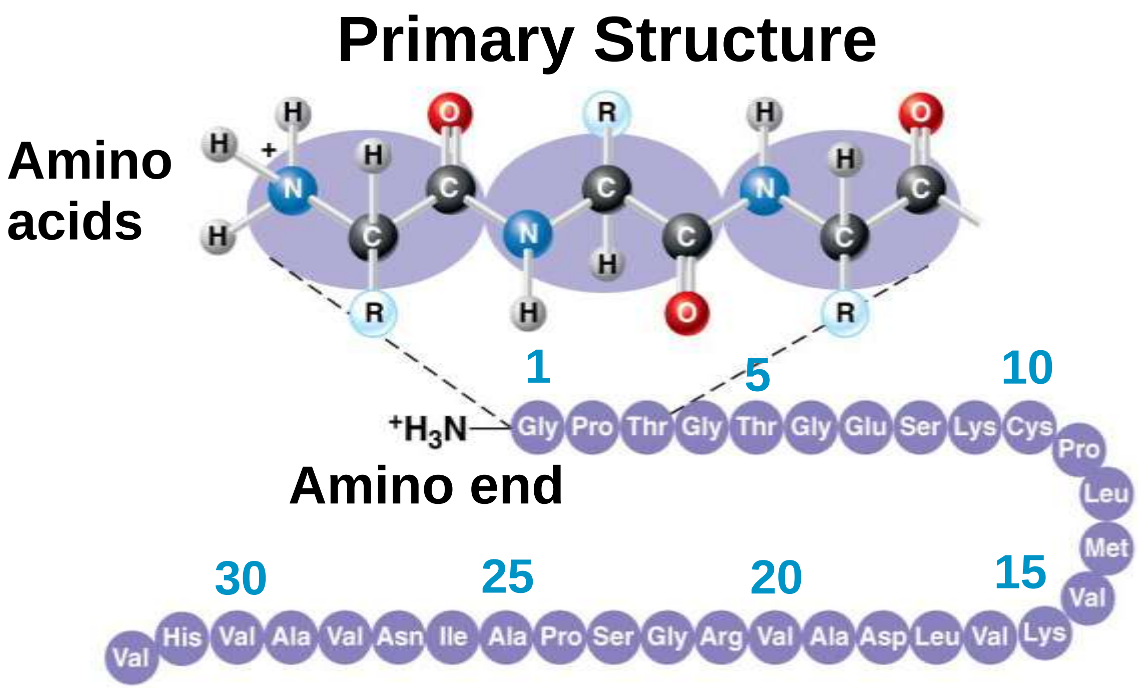

Secondary Structure
Tertiary Structure
Quaternary Structure
α helix
Hydrogen bond
β strand
Hydrogen bond
β pleated sheet
Transthyretin polypeptide
Transthyretin protein

a) The **primary structure** of a protein is its sequence of amino acids

b) Primary structure is like the order of letters in a long word

c) Primary structure is determined by inherited genetic information

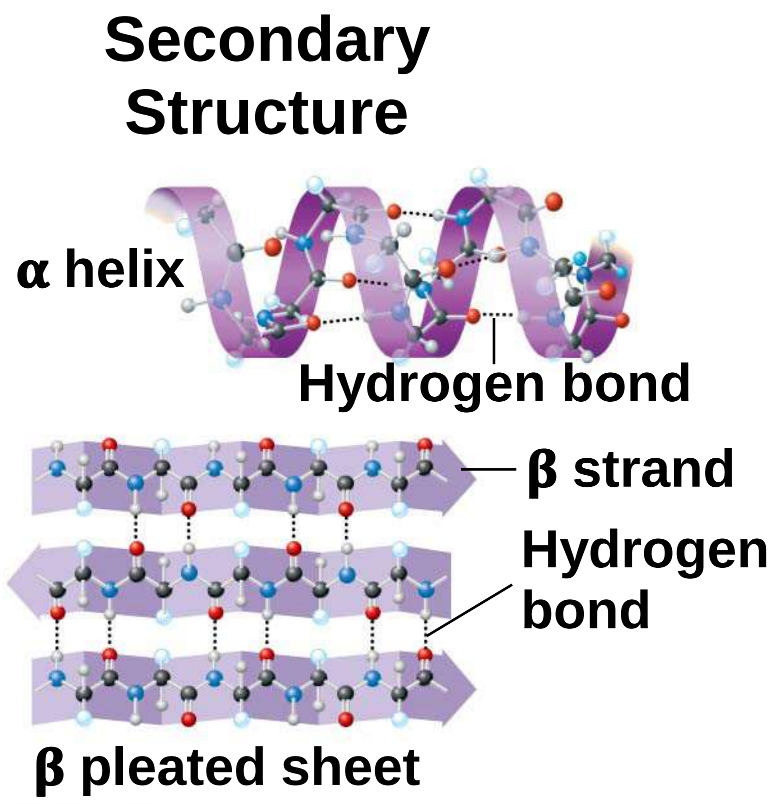
Figure 5.18aa



a) The coils and folds of **secondary structure** result from hydrogen bonds between repeating constituents of the polypeptide backbone

b) Typical secondary structures are a coil called an α **helix** and a folded structure called a β **pleated sheet**

Figure 5.18ba

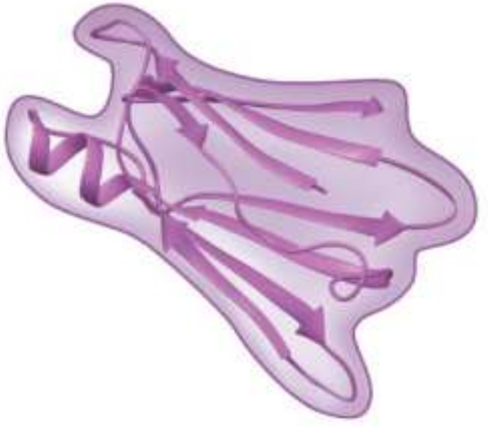


a) **Tertiary structure**, the overall shape of a polypeptide, results from interactions between R groups, rather than interactions between backbone constituents

b) These interactions include hydrogen bonds, ionic bonds, **hydrophobic interactions**, and van der Waals interactions

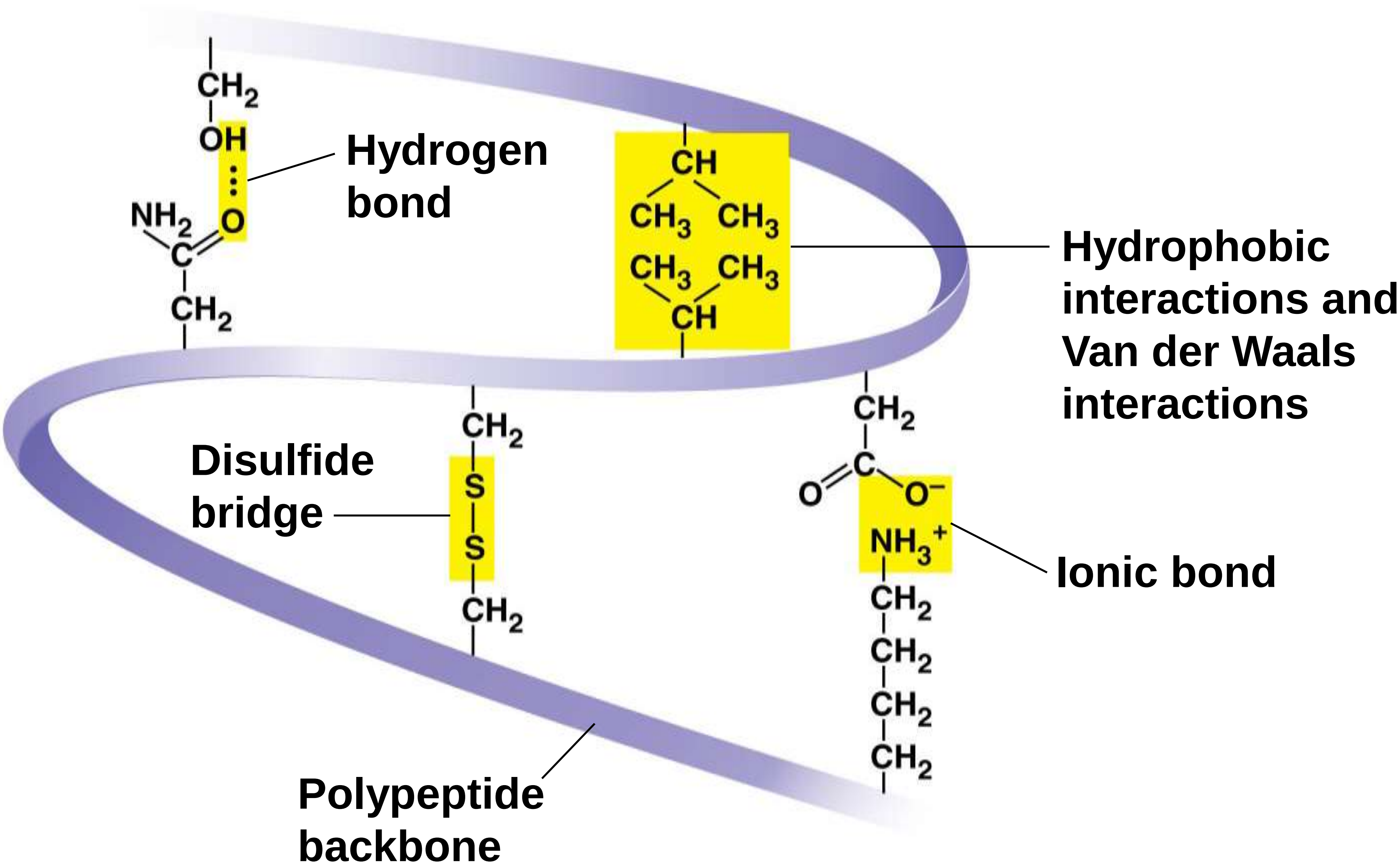
c) Strong covalent bonds called **disulfide** bridges may reinforce the protein's structure

Tertiary Structure



Transthyretin polypeptide

Figure 5.18d



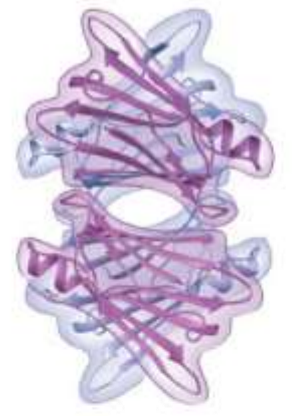
CH2
OH
Hydrogen bond
NH2
C
O
CH2
CH
CH3 CH3
CH3 CH3
CH
Hydrophobic interactions and Van der Waals interactions
CH2
Disulfide bridge
S
S
CH2
CH2
C
O
O–
NH3+
Ionic bond
CH2
CH2
CH2
CH2
Polypeptide backbone

a) **Quaternary structure** results when two or more polypeptide chains form one macromolecule

b) Collagen is a fibrous protein consisting of three polypeptides coiled like a rope

c) Hemoglobin is a globular protein consisting of four polypeptides: two alpha and two beta chains

Quaternary Structure



Transthyretin protein

Figure 5.18e

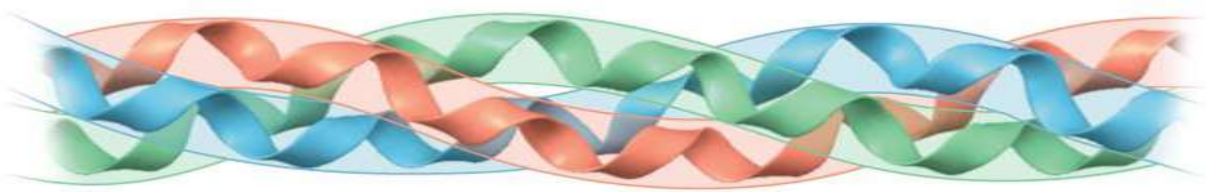
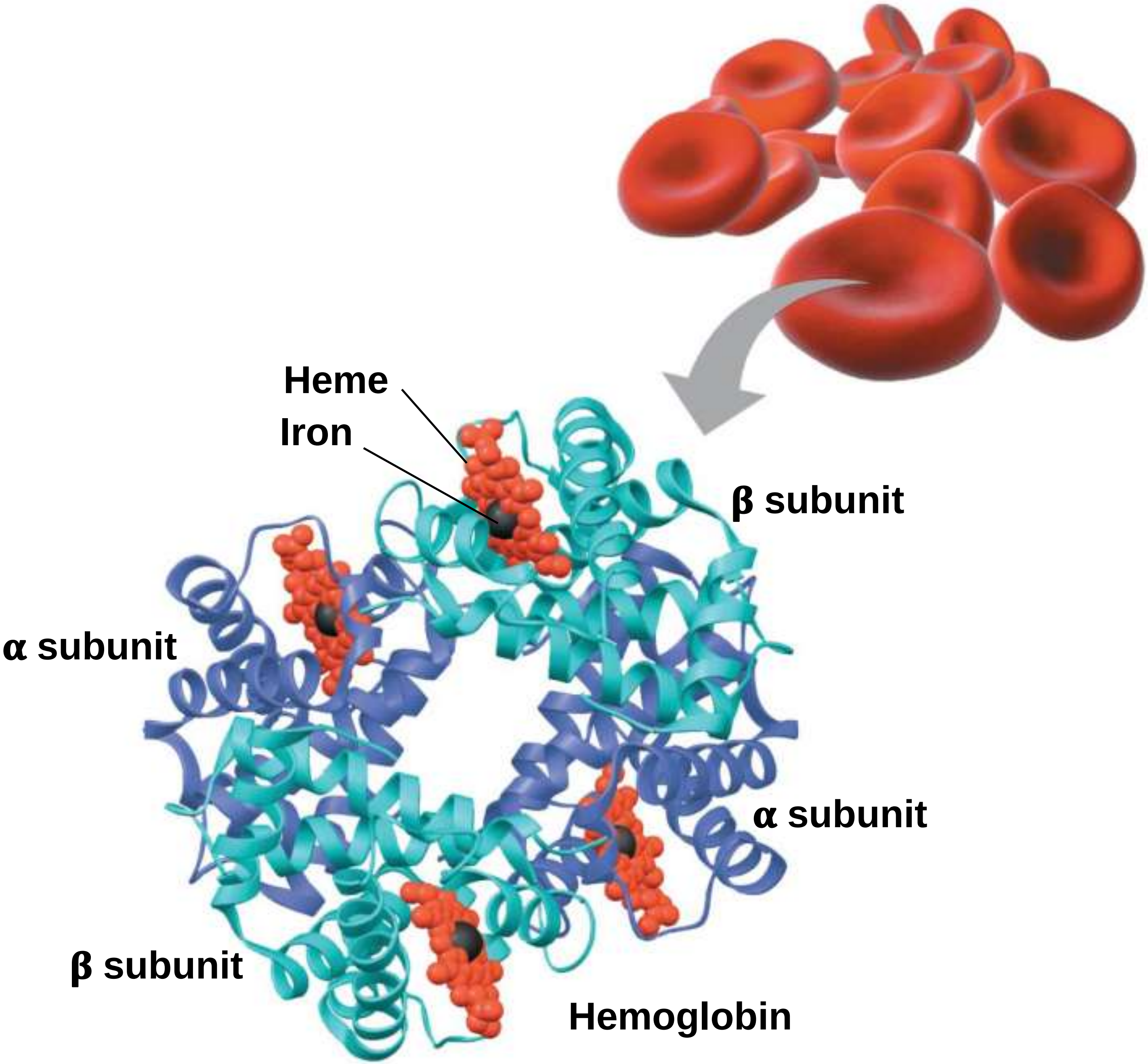


Figure 5.18f



Heme
Iron
β subunit
α subunit
α subunit
β subunit
Hemoglobin

Sickle-Cell Disease: A Change in Primary Structure

a) A slight change in primary structure can affect a protein’s structure and ability to function

b) Sickle-cell disease, an inherited blood disorder, results from a single amino acid substitution in the protein hemoglobin

Figure 5.19

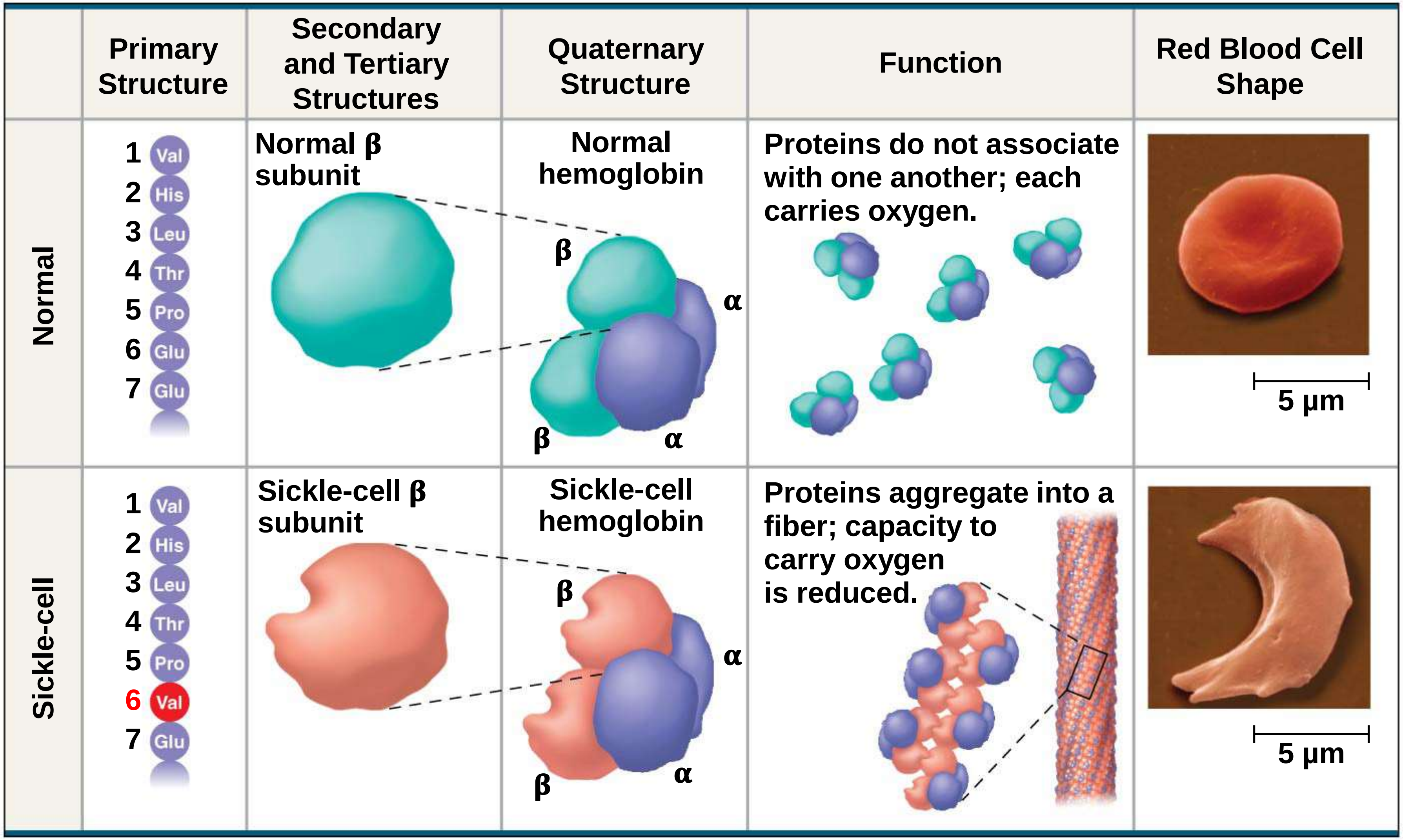

Primary Structure
Secondary and Tertiary Structures
Quaternary Structure
Function
Red Blood Cell Shape
Normal
1 Val
2 His
3 Leu
4 Thr
5 Pro
6 Glu
7 Glu
Normal β subunit
Normal hemoglobin
β
α
β
α
Proteins do not associate with one another; each carries oxygen.
5 µm
Sickle-cell
1 Val
2 His
3 Leu
4 Thr
5 Pro
6 Val
7 Glu
Sickle-cell β subunit
Sickle-cell hemoglobin
β
α
β
α
Proteins aggregate into a fiber; capacity to carry oxygen is reduced.
5 µm

Figure 5.19a

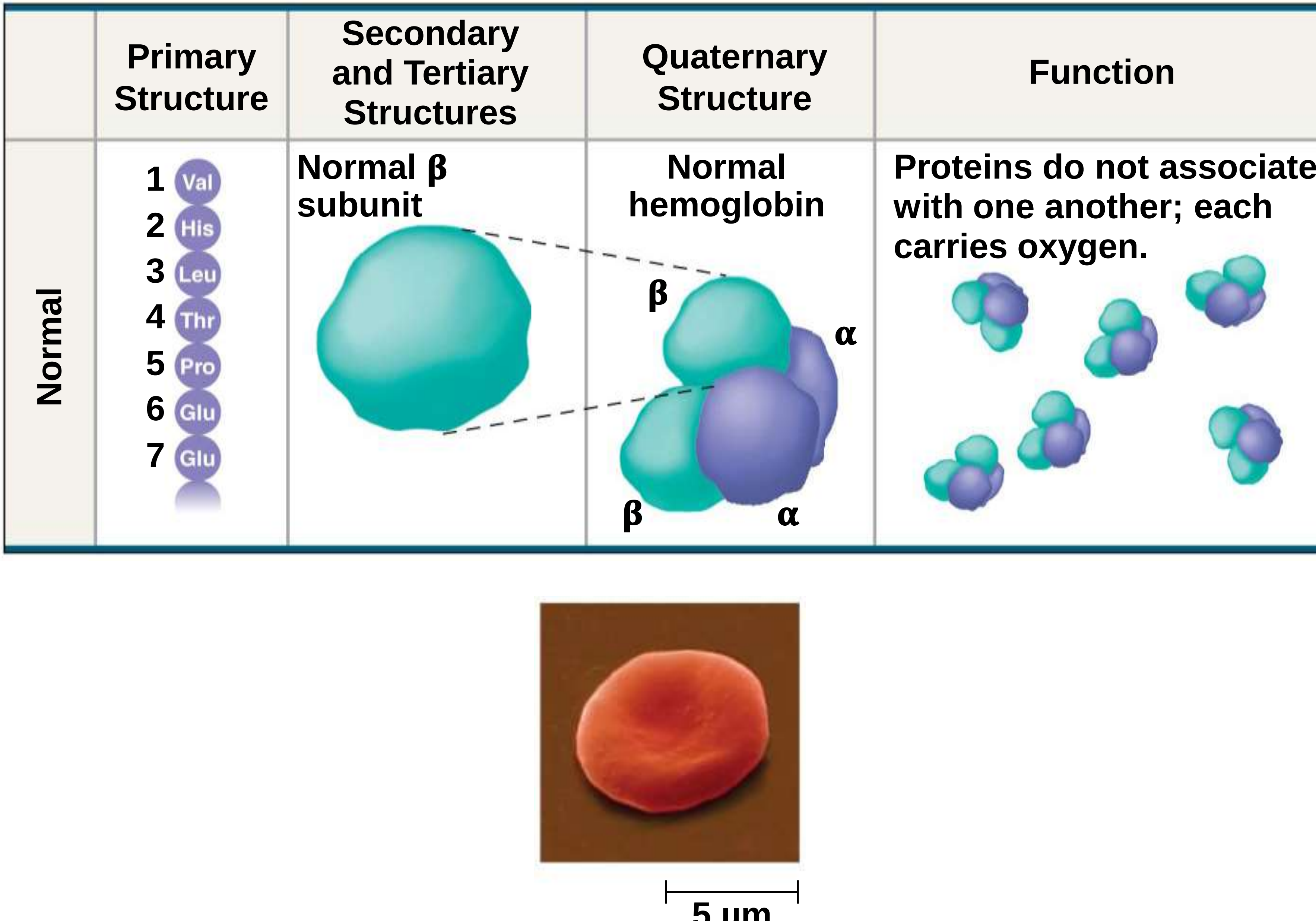
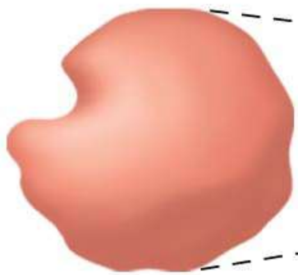
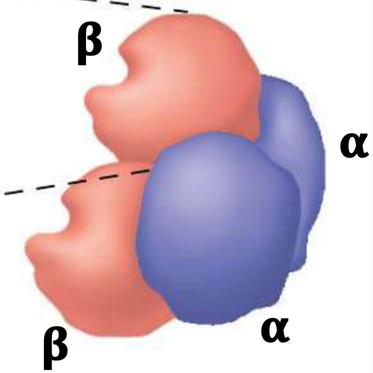
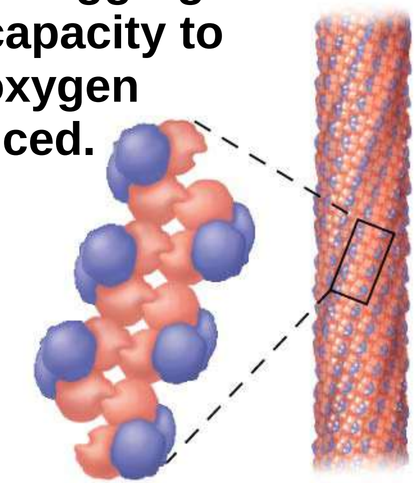

Primary Structure
Secondary and Tertiary Structures
Quaternary Structure
Function
Normal
1 Val
2 His
3 Leu
4 Thr
5 Pro
6 Glu
7 Glu
Normal β subunit
Normal hemoglobin
β
α
β
α
Proteins do not associate with one another; each carries oxygen.
5 µm

Figure 5.19b

	Primary Structure	Secondary and Tertiary Structures	Quaternary Structure	Function
Sickle-cell	1 Val 2 His 3 Leu 4 Thr 5 Pro 6 Val 7 Glu	Sickle-cell β subunit 	Sickle-cell hemoglobin 	Proteins aggregate into a fiber; capacity to carry oxygen is reduced. 

What Determines Protein Structure?

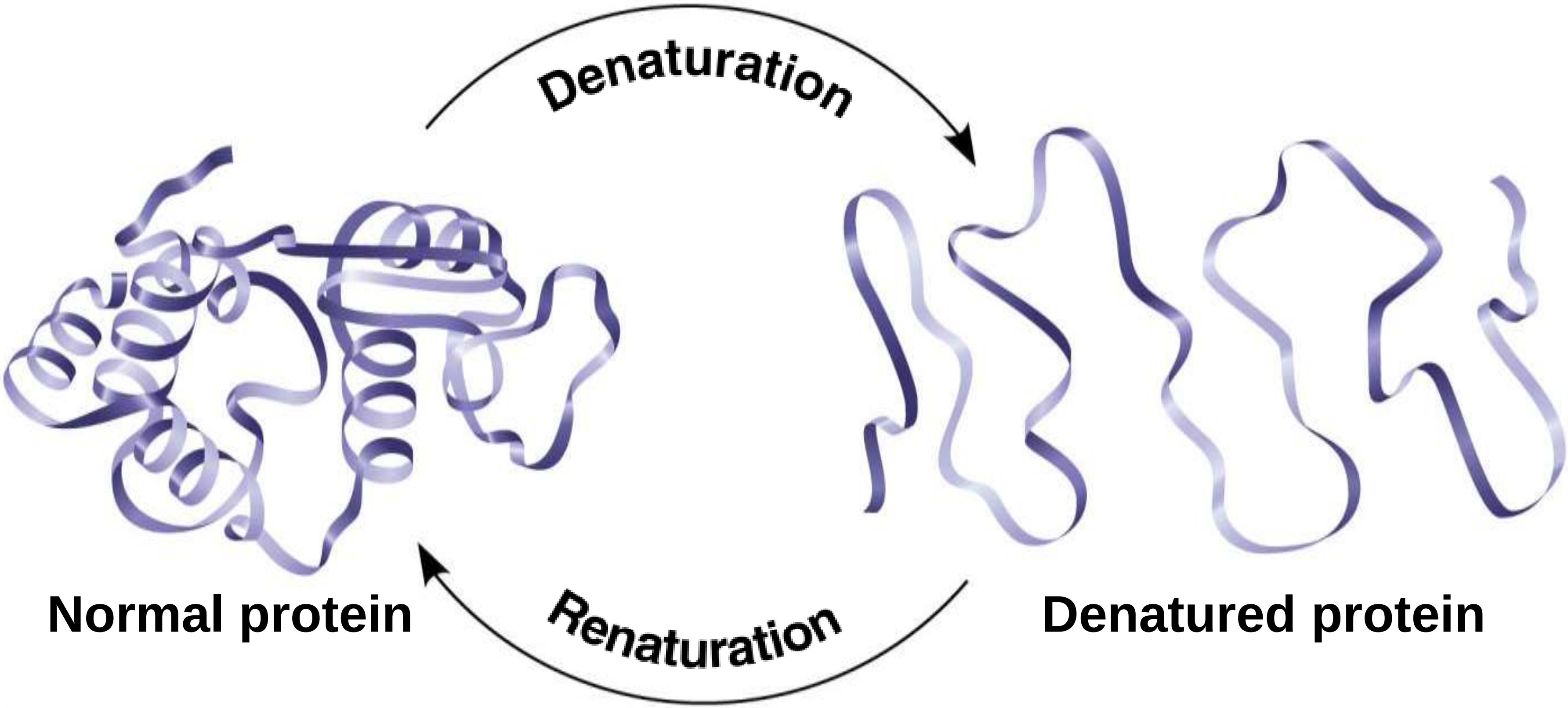
a) In addition to primary structure, physical and chemical conditions can affect structure

b) Alterations in pH, salt concentration, temperature, or other environmental factors can cause a protein to unravel

c) This loss of a protein’s native structure is called **denaturation**

d) A denatured protein is biologically inactive

Figure 5.20-3



Normal protein **Denatured protein**

Protein Folding in the Cell

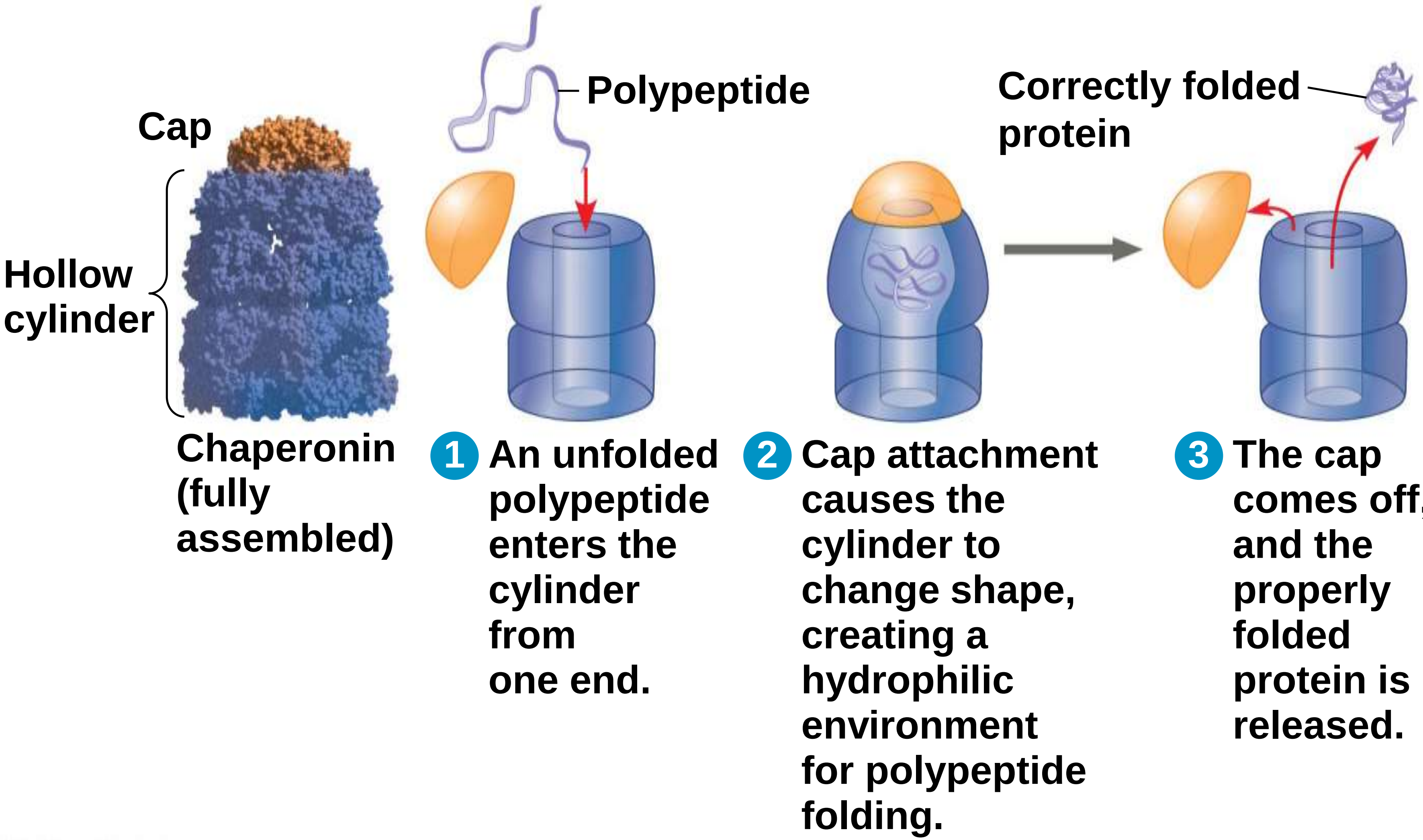
a) It is hard to predict a protein's structure from its primary structure

b) Most proteins probably go through several stages on their way to a stable structure

c) Chaperonins are protein molecules that assist the proper folding of other proteins

d) Diseases such as Alzheimer's, Parkinson's, and mad cow disease are associated with misfolded proteins

Figure 5.21



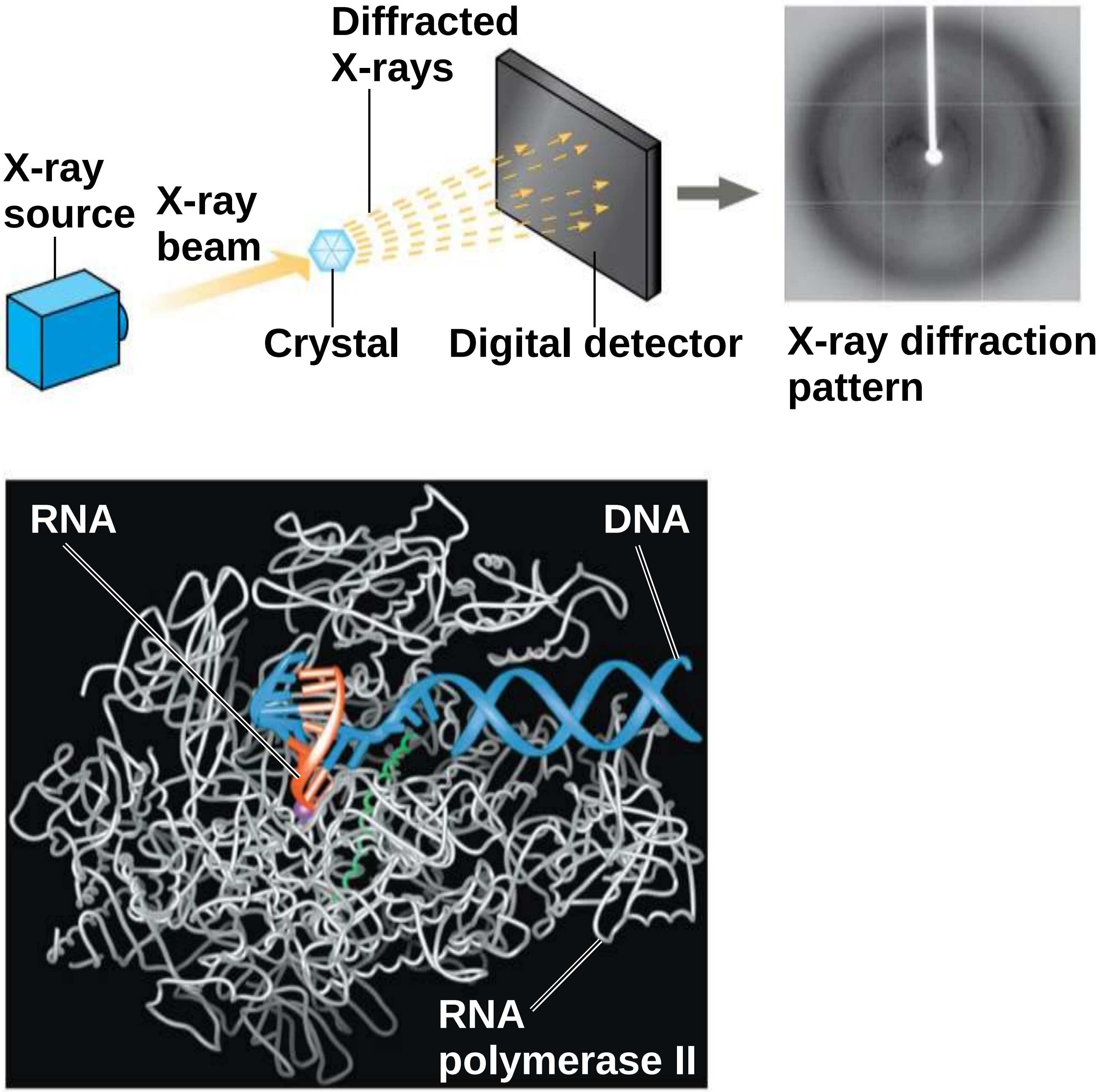
Cap
Hollow cylinder
Chaperonin (fully assembled)
Polypeptide
Correctly folded protein
1 An unfolded polypeptide enters the cylinder from one end.
2 Cap attachment causes the cylinder to change shape, creating a hydrophilic environment for polypeptide folding.
3 The cap comes off, and the properly folded protein is released.

a) Scientists use **X-ray crystallography** to determine a protein’s structure

b) Another method is nuclear magnetic resonance (NMR) spectroscopy, which does not require protein crystallization

c) Bioinformatics is another approach to prediction of protein structure from amino acid sequences

Figure 5.22



Diffracted X-rays
X-ray source
X-ray beam
Crystal
Digital detector
X-ray diffraction pattern
RNA
DNA
RNA polymerase II

Concept 5.5: Nucleic acids store, transmit, and help express hereditary information

a) The amino acid sequence of a polypeptide is programmed by a unit of inheritance called a **gene**

b) Genes consist of DNA, a **nucleic acid** made of monomers called nucleotides

The Roles of Nucleic Acids

a) There are two types of nucleic acids
    a) **Deoxyribonucleic acid (DNA)**
    b) **Ribonucleic acid (RNA)**
b) DNA provides directions for its own replication
c) DNA directs synthesis of messenger RNA (mRNA) and, through mRNA, controls protein synthesis
d) This process is called **gene expression**

Figure 5.23-1

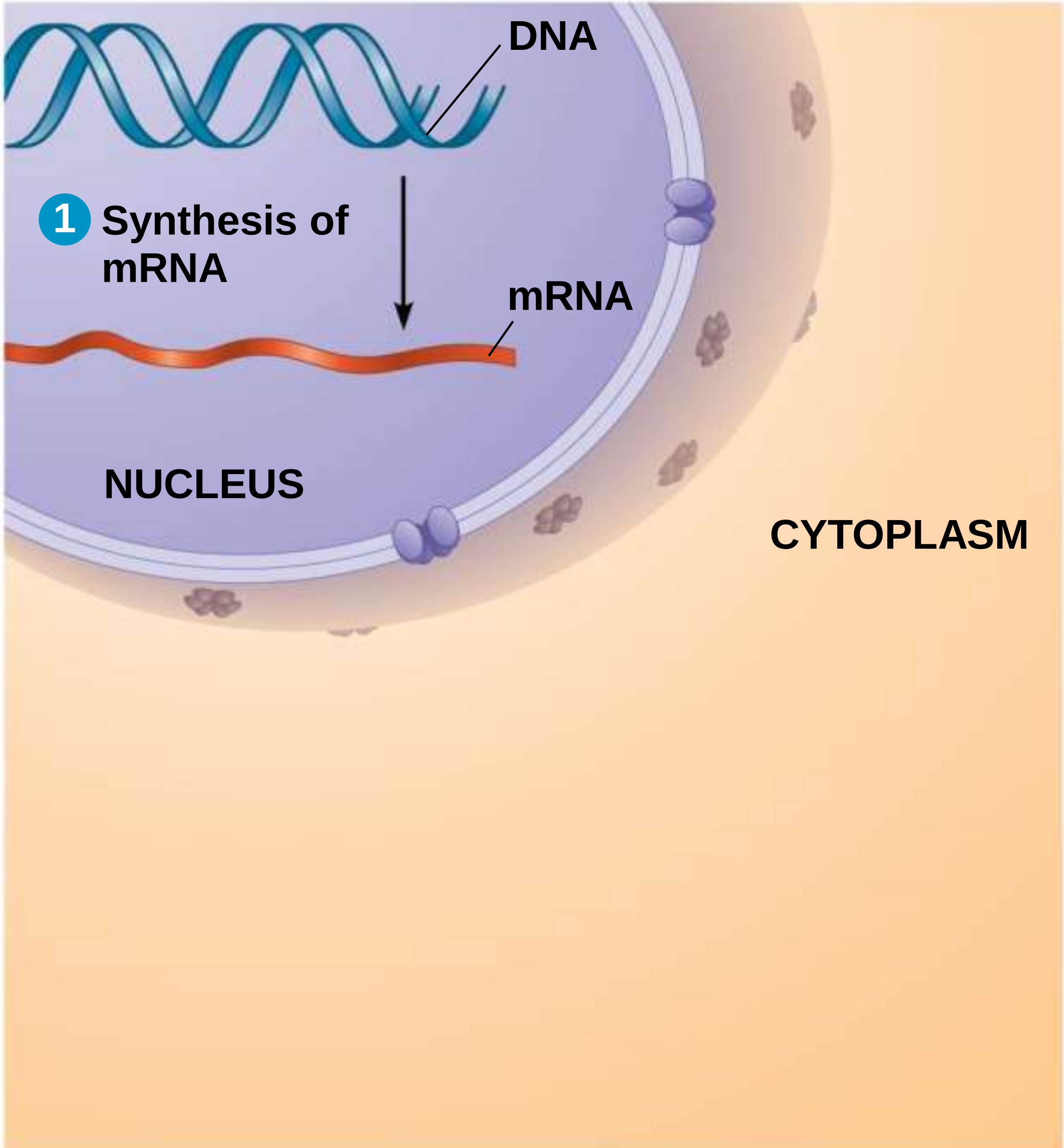
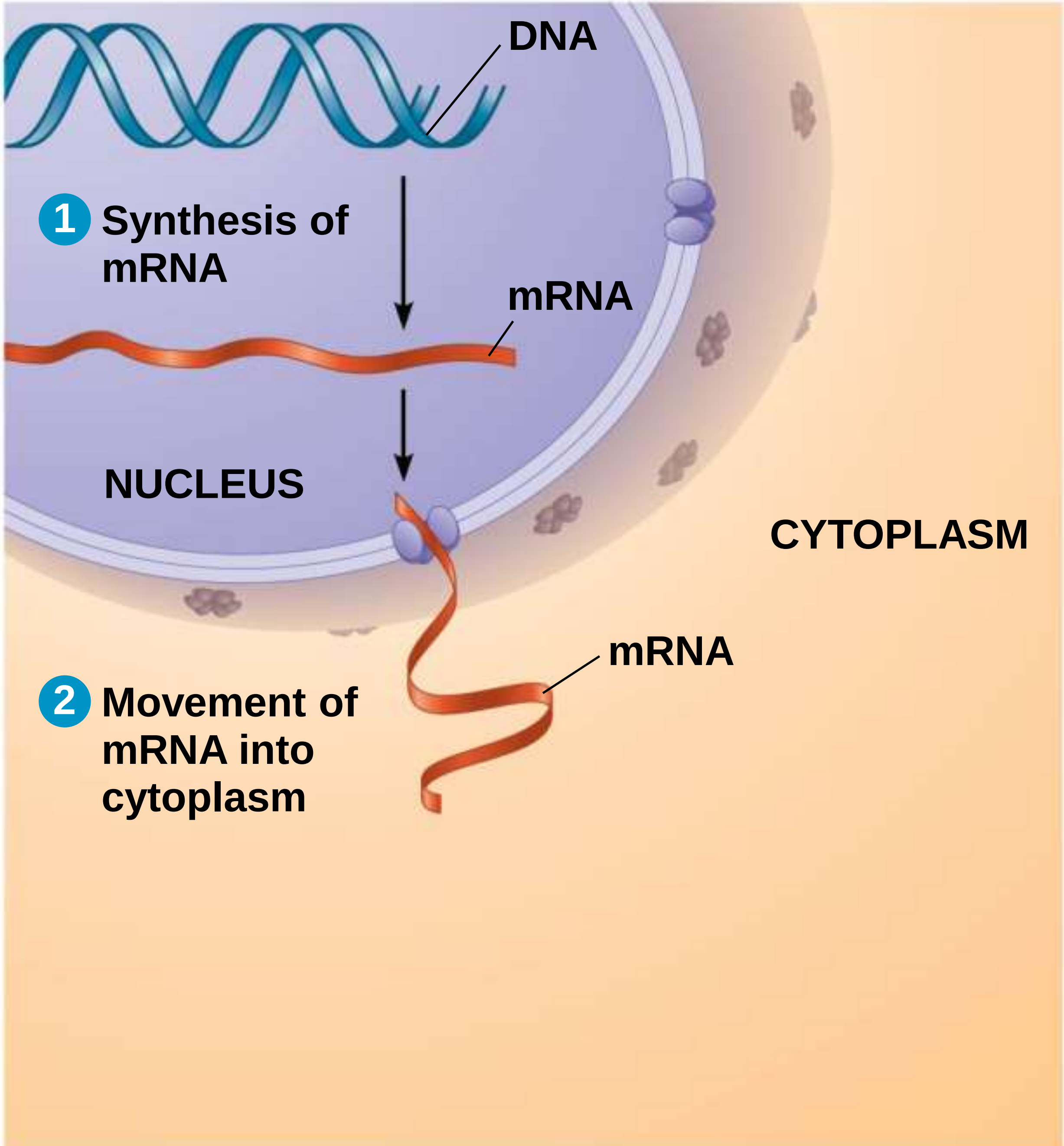
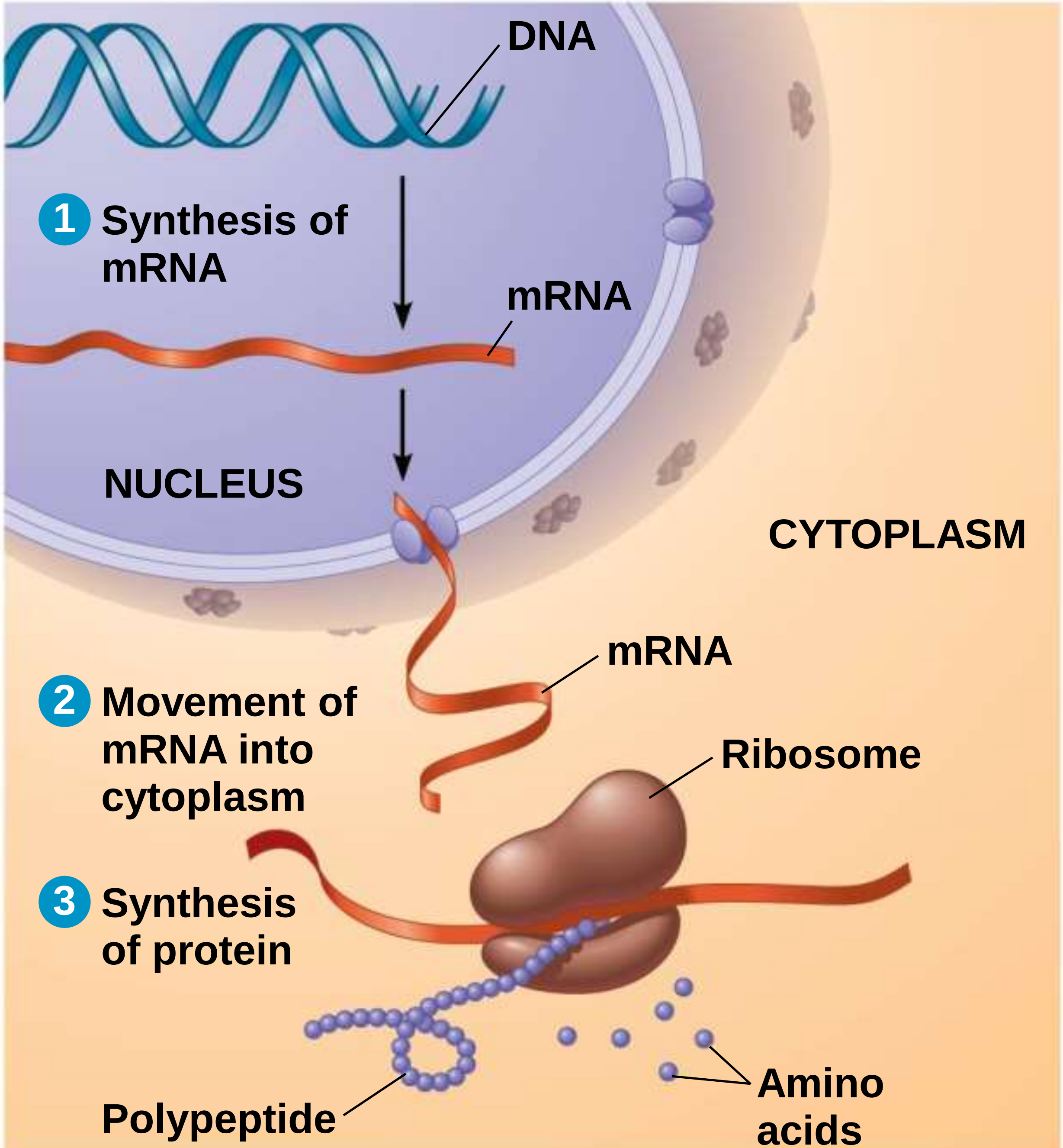


Figure 5.23-2



DNA
1 Synthesis of mRNA
mRNA
NUCLEUS
CYTOPLASM
mRNA
2 Movement of mRNA into cytoplasm

Figure 5.23-3


DNA
1 Synthesis of mRNA
mRNA
NUCLEUS
CYTOPLASM
mRNA
2 Movement of mRNA into cytoplasm
Ribosome
3 Synthesis of protein
Polypeptide
Amino acids

a) Each gene along a DNA molecule directs synthesis of a messenger RNA (mRNA)

b) The mRNA molecule interacts with the cell's protein-synthesizing machinery to direct production of a polypeptide

c) The flow of genetic information can be summarized as DNA → RNA → protein

The Components of Nucleic Acids

a) Nucleic acids are polymers called **polynucleotides**

b) Each polynucleotide is made of monomers called **nucleotides**

c) Each nucleotide consists of a nitrogenous base, a pentose sugar, and one or more phosphate groups

d) The portion of a nucleotide without the phosphate group is called a nucleoside

a) Nucleoside = nitrogenous base + sugar

b) There are two families of nitrogenous bases

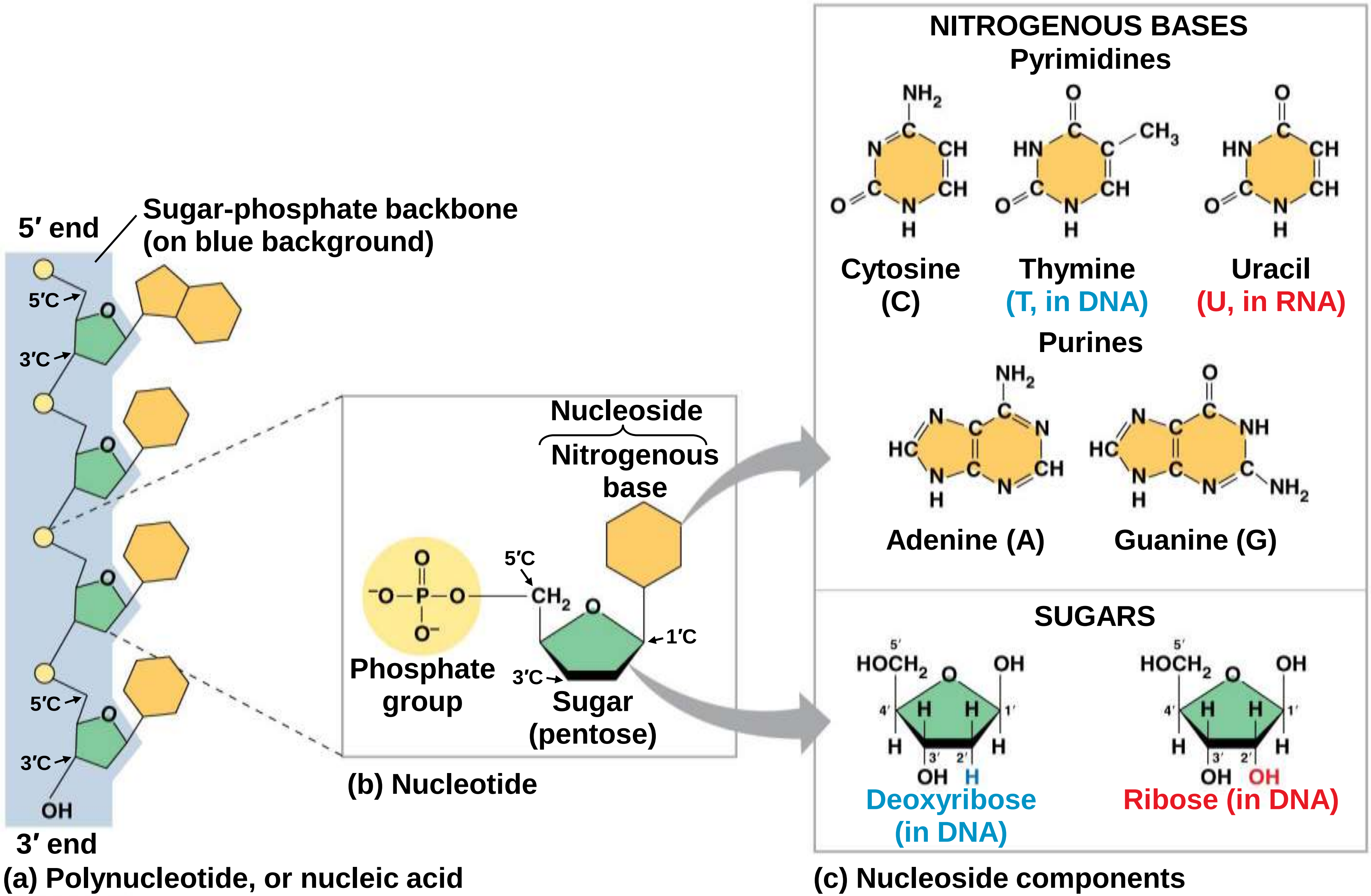
   a) **Pyrimidines** (cytosine, thymine, and uracil) have a single six-membered ring

   b) **Purines** (adenine and guanine) have a six-membered ring fused to a five-membered ring

c) In DNA, the sugar is **deoxyribose**; in RNA, the sugar is **ribose**

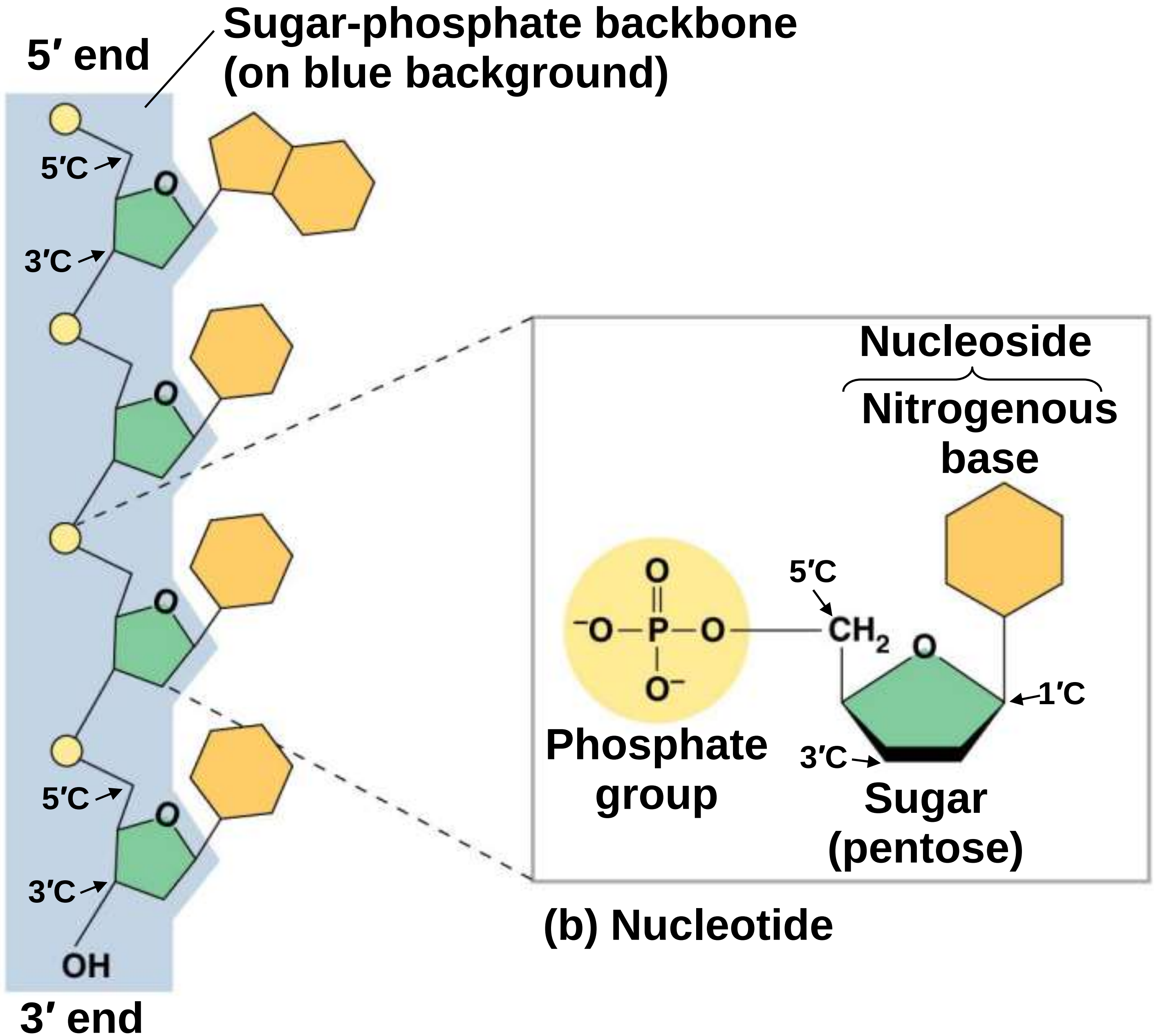
d) Nucleotide = nucleoside + phosphate group

Figure 5.24



5′ end
Sugar-phosphate backbone (on blue background)
5′C
3′C
O
OH
3′ end
(a) Polynucleotide, or nucleic acid
Nucleoside
Nitrogenous base
5′C
CH_2
O
1′C
3′C
$^-O-P-O$
O^-
Phosphate group
Sugar (pentose)
(b) Nucleotide
NITROGENOUS BASES
Pyrimidines
Cytosine (C)
Thymine (T, in DNA)
Uracil (U, in RNA)
Purines
Adenine (A)
Guanine (G)
SUGARS
$HOCH_2$
OH
Deoxyribose (in DNA)
Ribose (in DNA)
(c) Nucleoside components

Figure 5.24a

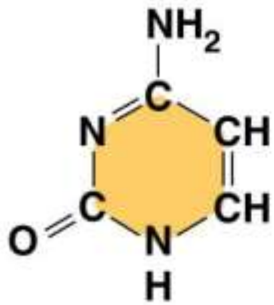


(a) Polynucleotide, or nucleic acid

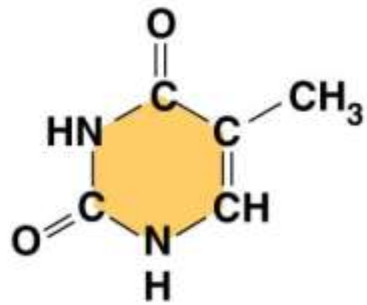
(b) Nucleotide

NITROGENOUS BASES

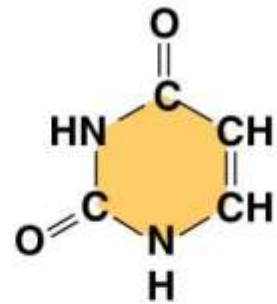
Pyrimidines



Cytosine (C)

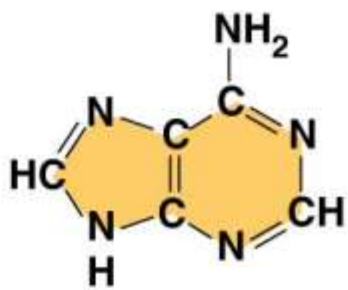


Thymine (T, in DNA)

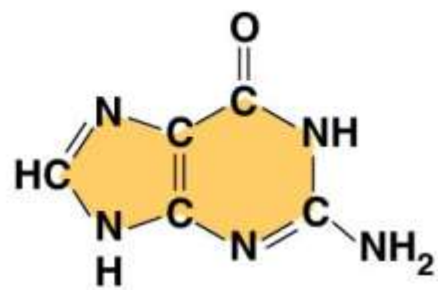


Uracil (U, in RNA)

Purines



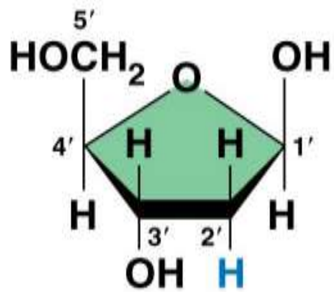
Adenine (A)



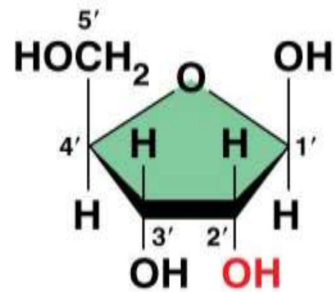
Guanine (G)

(c) Nucleoside components

SUGARS



Deoxyribose (in DNA)



Ribose (in DNA)

(c) Nucleoside components

Nucleotide Polymers

a) Nucleotides are linked together to build a polynucleotide

b) Adjacent nucleotides are joined by a phosphodiester linkage, which consists of a phosphate group that links the sugars of two nucleotides

c) These links create a backbone of sugar-phosphate units with nitrogenous bases as appendages

d) The sequence of bases along a DNA or mRNA polymer is unique for each gene

The Structures of DNA and RNA Molecules

a) DNA molecules have two polynucleotides spiraling around an imaginary axis, forming a **double helix**

b) The backbones run in opposite $5' \rightarrow 3'$ directions from each other, an arrangement referred to as **antiparallel**

c) One DNA molecule includes many genes

a) Only certain bases in DNA pair up and form hydrogen bonds: adenine (A) always with thymine (T), and guanine (G) always with cytosine (C)

b) This is called complementary base pairing

c) This feature of DNA structure makes it possible to generate two identical copies of each DNA molecule in a cell preparing to divide

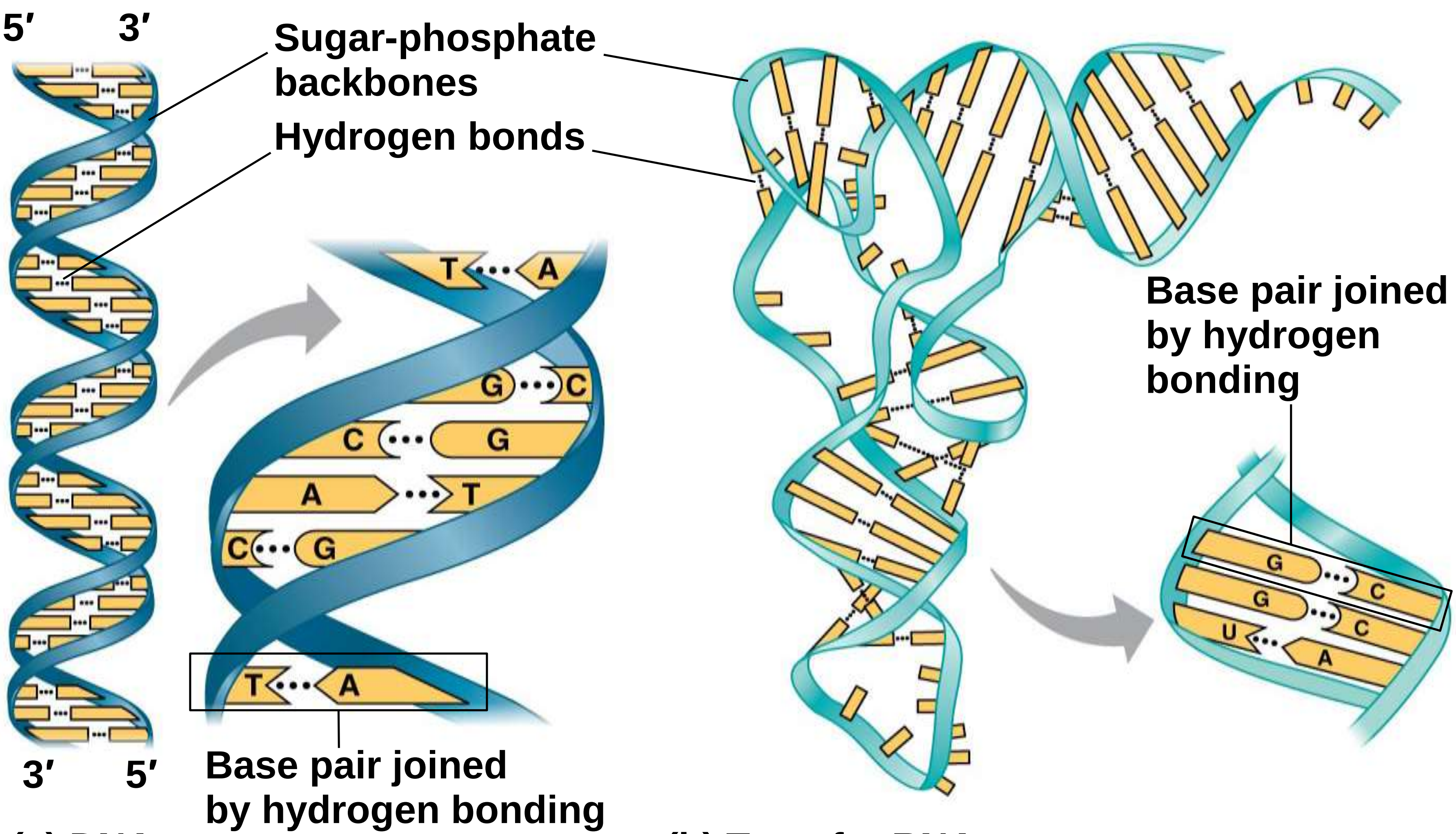
a) RNA, in contrast to DNA, is single stranded

b) Complementary pairing can also occur between two RNA molecules or between parts of the same molecule

c) In RNA, thymine is replaced by uracil (U) so A and U pair

d) While DNA always exists as a double helix, RNA molecules are more variable in form

Figure 5.25



(a) DNA

(b) Transfer RNA